

CALCULUS OF VARIATIONS

The calculus of variations deals with problems where we search for a function or curve, rather than a value of some variable, that makes a given quantity stationary, usually an energy or action integral. Because a function is varied, these problems are called **variational**. Variational principles, such as those of D'Alembert, Lagrange, and Hamilton, have been developed in classical mechanics; Fermat's principle (that of the shortest optical path) finds use in electrodynamics. Lagrangian variational techniques also occur in quantum mechanics and field theory. Before plunging into this rather different branch of mathematical physics, let us summarize some of its uses in both physics and mathematics.

1. **In existing physical theories:**
 - a. Unification of diverse areas of physics using energy as a key concept
 - b. Convenience in analysis: Lagrange equations, [Section 22.2](#)
 - c. Elegant treatment of constraints, [Section 22.4](#)
2. **Starting point for new, complex areas of physics and engineering.** In general relativity, the geodesic is taken as the minimum path of a light pulse or the free-fall path of a particle in curved Riemannian space. Variational principles appear in quantum field theory. Variational principles have been applied extensively in control theory.
3. **Mathematical unification.** Variational analysis provides a proof of the completeness of the Sturm-Liouville eigenfunctions, and can be used to establish bounds for the eigenvalues. Similar results follow for the eigenvalues and eigenfunctions in the Hilbert-Schmidt theory of integral equations.

22.1 EULER EQUATION

The calculus of variations typically involves problems in which a quantity to be minimized (or maximized) appears as a **functional**, meaning that it is a quantity whose argument(s) are themselves function(s), not just variable(s). As a simple, yet fairly general case, let J

be a functional of y , defined as

$$J[y] = \int_{x_1}^{x_2} f\left(y(x), \frac{dy(x)}{dx}, x\right) dx. \quad (22.1)$$

Here f is a fixed function of the three variables y , dy/dx , and x , while J will have a value dependent on the choice of y . The square-bracket notation is frequently used to remind the reader that J is a functional. Because J is given as an integral, its value depends on the behavior of $y(x)$ throughout the entire range of x (here $x_1 \leq x \leq x_2$). A typical problem in the calculus of variations is to find (usually subject to some constraints) a continuous and differentiable function $y(x)$ that makes J stationary relative to small changes in y anywhere (or everywhere) in its range of definition. These stationary values of J will in many problems be minima or maxima, but they can also be saddle points. The conditions of physical problems will normally require that variations in y be restricted to those that preserve its continuity and differentiability

It is convenient to introduce a notation that makes our discussions less cumbersome; we usually rewrite Eq. (22.1) in a notation with dy/dx denoted y_x and with the arguments x and $[y]$ suppressed, and we indicate the variation in J produced by a (small) variation in y as

$$\delta J = \delta \int_{x_1}^{x_2} f(y, y_x, x) dx. \quad (22.2)$$

Note that we wrote δ rather than d or ∂ ; this distinction reminds us that the variation is that of a function (here y) rather than that of a variable.

In visualizing the situation described by Eq. (22.2), it is helpful to think of $y(x)$ as a **path** or curve connecting the values $y(x_1)$ and $y(x_2)$; in fact, a common problem in the calculus of variations will be to determine $y(x)$ subject to the constraint that $y(x_1)$ and $y(x_2)$ have specified values (and often subject to further constraints that may also be integrals). To illustrate the class of problems represented by Eq. (22.2), here are two simple examples:

- Determination of the minimum-energy configuration of a rope or chain of given length attached to fixed points at both ends, in the presence of a uniform gravitational field.
- Determination of the track between two points at different heights that will minimize the travel time of an object that, starting from rest, slides without friction along the track subject only to a uniform gravitational field (this is known as the **brachistochrone problem**).

The problems here under consideration are much more difficult than typical minimizations in differential calculus, where the minimum in a function can be found by comparing its values, say $y(x)$, at neighboring points (by looking at dy/dx). What we can do, instead, is to start by assuming the existence of an optimum path, i.e., a function $y(x)$ for which J is stationary, and then compare J for our (unknown) optimum path with that obtained from neighboring paths, of which there are an infinite number. See Fig. 22.1. Even this strategy may sometimes fail, as there exist functionals J for which there is no optimum path.

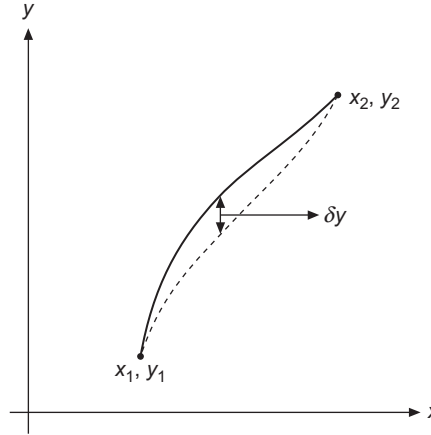


FIGURE 22.1 Neighboring paths.

Restricting attention to functions $y(x)$ for which the endpoints $y(x_1)$ and $y(x_2)$ are fixed, we consider a deformation of $y(x)$, called the **variation** of y and denoted δy . We describe δy by introducing a new function, $\eta(x)$, and a scale factor α that controls the magnitude of the variation. The function $\eta(x)$ is arbitrary except for being continuous and differentiable, and, to keep the endpoints fixed, with

$$\eta(x_1) = \eta(x_2) = 0. \quad (22.3)$$

With these definitions, our path, now a function of α , is

$$y(x, \alpha) = y(x, 0) + \alpha \eta(x), \quad (22.4)$$

and we choose $y(x, 0)$ as the (unknown) path that will minimize J . Relative to $y(x, 0)$, the variation δy is then

$$\delta y = \alpha \eta(x). \quad (22.5)$$

Using Eq. (22.4), our formula for J can now be written

$$J(\alpha) = \int_{x_1}^{x_2} f(y(x, \alpha), y_x(x, \alpha), x) dx, \quad (22.6)$$

and we see that we have reached a simpler formulation in which J is now a **function** of α rather than a **functional** of y . This means that we now know how to optimize it.¹

We proceed now to obtain a stationary value of J by imposing the condition

$$\left[\frac{\partial J(\alpha)}{\partial \alpha} \right]_{\alpha=0} = 0, \quad (22.7)$$

analogous to the vanishing of the derivative dy/dx in differential calculus.

¹The arbitrary nature of the dependence of $J(\alpha)$ on $\eta(x)$ will come into play later.

Now, the α dependence of the integral is contained in $y(x, \alpha)$ and $y_x(x, \alpha) = (\partial/\partial x)y(x, \alpha)$. Therefore,²

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} \frac{\partial y}{\partial \alpha} + \frac{\partial f}{\partial y_x} \frac{\partial y_x}{\partial \alpha} \right] dx = 0. \quad (22.8)$$

From Eq. (22.4),

$$\frac{\partial y(x, \alpha)}{\partial \alpha} = \eta(x) \quad \text{and} \quad \frac{\partial y_x(x, \alpha)}{\partial \alpha} = \frac{d\eta(x)}{dx}, \quad (22.9)$$

so Eq. (22.8) becomes

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y_x} \frac{d\eta(x)}{dx} \right) dx = 0. \quad (22.10)$$

Integrating the second term by parts to get $\eta(x)$ as a common factor, we convert it to

$$\int_{x_1}^{x_2} \frac{d\eta(x)}{dx} \frac{\partial f}{\partial y_x} dx = \eta(x) \frac{\partial f}{\partial y_x} \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \eta(x) \frac{d}{dx} \frac{\partial f}{\partial y_x} dx. \quad (22.11)$$

The integrated part vanishes by Eq. (22.3), and Eq. (22.10) becomes

$$\frac{\partial J(\alpha)}{\partial \alpha} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y_x} \right] \eta(x) dx = 0. \quad (22.12)$$

Equation (22.12), which must be satisfied for arbitrary $\eta(x)$, is to be understood as a condition on $y(x)$. Occasionally we will see Eq. (22.12) multiplied by $\delta\alpha$, which gives, upon using $\eta(x)\delta\alpha = \delta y$,

$$\delta J = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y_x} \right) \delta y dx = 0. \quad (22.13)$$

Equation (22.13) is to be solved for arbitrary δy with $\delta y(x_1) = \delta y(x_2) = 0$.

We now take up the solution of Eq. (22.12). That equation can be satisfied for arbitrary $\eta(x)$ only if the bracketed expression forming the remainder of its integrand vanishes “almost everywhere,” meaning everywhere except possibly at isolated points.³ The condition for our stationary value is thus formally a partial differential equation (PDE),

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y_x} = 0, \quad (22.14)$$

known as the **Euler equation**. Since the form of f is known, it will actually reduce (because there is really only one independent variable, x) to an ordinary differential equation (ODE) for y with boundary conditions at x_1 and x_2 . In that connection, it is important

²Note that y and y_x are being treated as **independent** variables because they occur as different arguments of f .

³Compare the discussion of convergence in the mean, at Eq. (5.22).

to note that the derivative d/dx occurs in the Euler equation, and that it has a meaning distinct from the partial derivative $\partial/\partial x$. In particular, if $f = f(y(x), y_x, x)$, then df/dx , which stands for the change in f (from all sources) due to a change in x , has the evaluation

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y_x} \frac{d^2y}{dx^2},$$

where the last term has the form given because $dy_x/dx = d^2y/dx^2$. Note that the first term on the right gives the **explicit** x -dependence of f ; the second and third terms give its **implicit** x -dependence via y and y_x .

The Euler equation, Eq. (22.14), is a necessary, but by no means sufficient condition that there be a function $y(x)$ that is continuous and differentiable on the range (x_1, x_2) and yields a stationary value of J .⁴ A nice example of a lack of sufficiency is provided by the problem of determining stationary paths between points on the surface of a sphere (this example was provided by Courant and Robbins; see Additional Readings). The minimum-distance path from point A to point B on a spherical surface is the arc of a great circle, shown as Path 1 in Fig. 22.2. But Path 2 also satisfies the Euler equation. Path 2 is a maximum, but only if we demand that it be a great circle and then only if we make less than one circuit (as Path 2 plus n complete revolutions is also a solution). If the path is not required to be a great circle, any deviation from Path 2 will increase the length. This is hardly the property of a local maximum, and that illustrates why it is important to check solutions of the Euler equation to see if they satisfy the physical conditions of the given problem.

Sometimes a problem admits a discontinuous solution that has physical relevance and will not be found by straightforward application of the Euler equation. An example is provided by the soap film of Example 22.1.3, where such a solution describes what happens if the film becomes unstable and breaks.

Following are examples of the use of the Euler equation.

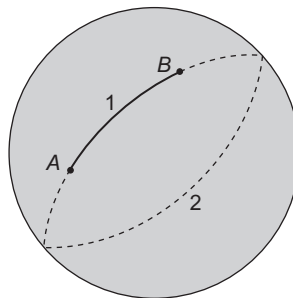


FIGURE 22.2 Stationary paths over a sphere.

⁴For a discussion of sufficiency conditions and the development of the calculus of variations as a part of mathematics, see the works by Ewing and Sagan in Additional Readings.

Example 22.1.1 STRAIGHT LINE

Perhaps the simplest application of the Euler equation is in the determination of the shortest distance between two points in the Euclidean xy -plane. Since the element of distance is

$$ds = [(dx)^2 + (dy)^2]^{1/2} = [1 + y_x^2]^{1/2} dx,$$

the distance J may be written as

$$J = \int_{x_1, y_1}^{x_2, y_2} ds = \int_{x_1}^{x_2} [1 + y_x^2]^{1/2} dx. \quad (22.15)$$

Comparison with Eq. (22.2) shows that

$$f(y, y_x, x) = (1 + y_x^2)^{1/2}.$$

Substituting into Eq. (22.14) and noting that $\partial f / \partial y$ vanishes, we obtain

$$-\frac{d}{dx} \left[\frac{1}{(1 + y_x^2)^{1/2}} \right] = 0,$$

or

$$\frac{1}{(1 + y_x^2)^{1/2}} = C, \quad \text{a constant.}$$

This equation is satisfied if

$$y_x = a, \quad \text{a second constant.}$$

Integrating this expression for y_x , we get

$$y = ax + b, \quad (22.16)$$

which is the familiar equation for a straight line. The constants a and b are now chosen so that the line passes through the two points (x_1, y_1) and (x_2, y_2) . Hence the Euler equation predicts that the shortest⁵ distance between two fixed points in Euclidean space is a straight line. ■

The generalization of this to curved four-dimensional space-time leads to the important concept of the geodesic in general relativity. A further discussion of geodesics is in Section 22.2.

⁵Technically, we have only found a $y(x)$ of stationary J . By inspection of the solution, we easily determine the distance to be a minimum.

Example 22.1.2 OPTICAL PATH NEAR A BLACK HOLE

We now wish to determine the optical path in an atmosphere where the velocity of light increases in proportion to the height y according to $v(y) = y/b$, with $b > 0$ some parameter describing the light speed. So $v = 0$ at $y = 0$, which simulates the conditions at the surface of a black hole, called its **event horizon**, where the gravitational force is so strong that the velocity of light goes to zero, thus trapping light.

Our variational principle (Fermat's principle) is that light will take the path of shortest travel time from (x_1, y_1) to (x_2, y_2) , namely

$$\Delta t = \int dt = \int_{x_1, y_1}^{x_2, y_2} \frac{ds}{v} = \int_{x_1, y_1}^{x_2, y_2} \frac{b}{y} ds = b \int_{x_1, y_1}^{x_2, y_2} \frac{\sqrt{dx^2 + dy^2}}{y} = \text{minimum.} \quad (22.17)$$

The path is along a line defined by the relation between y and x . While we have in previous equations taken x to be the independent variable, there is no inherent requirement to do so, and our work on the present problem will be simplified if we choose y as the independent variable, and we write Eq. (22.17) in the form

$$\Delta t = \int_{y_1}^{y_2} \frac{\sqrt{x_y^2 + 1}}{y} dy, \quad (22.18)$$

where x_y stands for dx/dy . Then our Euler equation will be

$$\frac{\partial f}{\partial x} - \frac{d}{dy} \frac{\partial f}{\partial x_y} = 0, \quad \text{with} \quad f(x, x_y, y) = \frac{\sqrt{x_y^2 + 1}}{y}.$$

Noting that $\partial f / \partial x = 0$ and differentiating $\partial f / \partial x_y$, we have

$$-\frac{d}{dy} \frac{x_y}{y\sqrt{x_y^2 + 1}} = 0.$$

This equation can be integrated, giving

$$\frac{x_y}{y\sqrt{x_y^2 + 1}} = C_1 = \text{constant}, \quad \text{or} \quad x_y = \frac{C_1 y}{\sqrt{1 - C_1^2 y^2}}.$$

Writing $x_y = dx/dy$ and separating dx and dy in this first-order ODE, we find the integral

$$\int dx = \int \frac{C_1 y dy}{\sqrt{1 - C_1^2 y^2}},$$

which yields

$$x + C_2 = -\frac{\sqrt{1 - C_1^2 y^2}}{C_1}, \quad \text{or} \quad (x + C_2)^2 + y^2 = \frac{1}{C_1^2}.$$

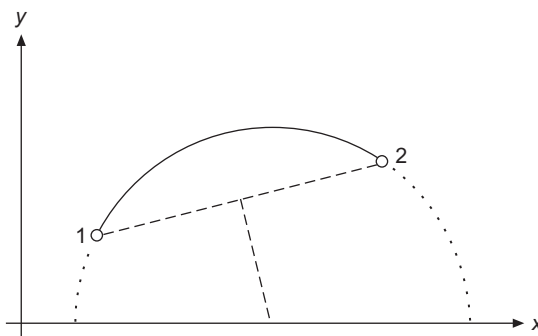


FIGURE 22.3 Circular optical path in medium.

Irrespective of the values of C_1 and C_2 , this light path is the arc of a circle whose center is on the line $y = 0$, namely the event horizon. The actual path of light passing from (x_1, y_1) to (x_2, y_2) will be on the circle through those points centered on $y = 0$; the construction of the path can be performed geometrically as shown in Fig. 22.3. Note that light will not escape completely from the black hole with this model for $v(y)$ unless $x_1 = x_2$ (a path perpendicular to the event horizon).

This example may be adapted to a mirage (Fata Morgana) in a desert with hot air near the ground and cooler air aloft (the index of refraction changes with height in cool vs. hot air). For the mirage problem, the relevant velocity law is $v(y) = v_0 - y/b$. In that case, the circular light path is no longer convex with center on the x -axis, but becomes concave. ■

Alternate Forms of Euler Equations

Another form of the Euler equation, which is often useful (Exercise 22.1.1), is

$$\frac{\partial f}{\partial x} - \frac{d}{dx} \left(f - y_x \frac{\partial f}{\partial y_x} \right) = 0. \quad (22.19)$$

In problems in which $f = f(y, y_x)$, i.e., in which x does not appear explicitly, Eq. (22.19) reduces to

$$\frac{d}{dx} \left(f - y_x \frac{\partial f}{\partial y_x} \right) = 0, \quad (22.20)$$

or

$$f - y_x \frac{\partial f}{\partial y_x} = \text{constant}. \quad (22.21)$$

Example 22.1.3 SOAP FILM

As our next illustrative example, consider two parallel coaxial wire circles to be connected by a surface of minimum area that is generated by revolving a curve $y(x)$ about the x -axis.

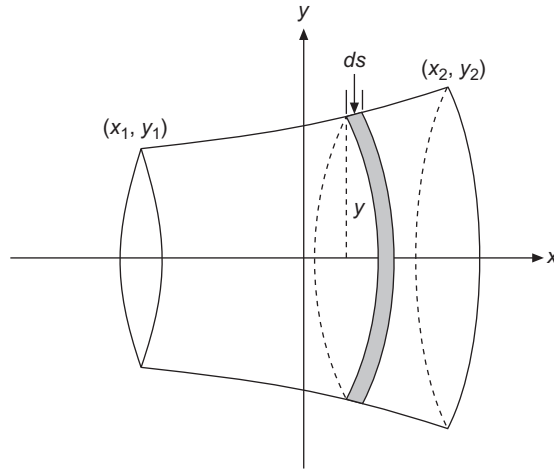


FIGURE 22.4 Surface of rotation, soap-film problem.

See Fig. 22.4. The curve is required to pass through fixed endpoints (x_1, y_1) and (x_2, y_2) . The variational problem is to choose the curve $y(x)$ so that the area of the resulting surface will be a minimum. A physical situation corresponding to this problem is that of a soap film suspended between the wire circles.

For the element of area shown in Fig. 22.4,

$$dA = 2\pi y \, ds = 2\pi y(1 + y_x^2)^{1/2} dx.$$

The variational equation is then

$$J = \int_{x_1}^{x_2} 2\pi y(1 + y_x^2)^{1/2} dx.$$

Neglecting the 2π , we identify

$$f(y, y_x, x) = y(1 + y_x^2)^{1/2}.$$

Since $\partial f / \partial x = 0$, we may apply Eq. (22.20) and get

$$y(1 + y_x^2)^{1/2} - \frac{y y_x^2}{(1 + y_x^2)^{1/2}} = c_1,$$

which simplifies to

$$\frac{y}{(1 + y_x^2)^{1/2}} = c_1. \quad (22.22)$$

Squaring, we get

$$\frac{y^2}{1 + y_x^2} = c_1^2,$$

which rearranges to

$$(y_x)^{-1} = \frac{dx}{dy} = \frac{c_1}{\sqrt{y^2 - c_1^2}}. \quad (22.23)$$

We note in passing that c_1 had better have a value that causes dy/dx to be real. Equation (22.23) may be integrated to give

$$x = c_1 \cosh^{-1} \frac{y}{c_1} + c_2,$$

and, solving for y , we have

$$y = c_1 \cosh \left(\frac{x - c_2}{c_1} \right). \quad (22.24)$$

Finally, c_1 and c_2 are determined by requiring the solution to pass through the points (x_1, y_1) and (x_2, y_2) . Our “minimum”-area surface is a special case of a catenary of revolution, or a **catenoid**. ■

Soap Film: Minimum Area

This calculus of variations contains many pitfalls for the unwary. Remember, the Euler equation is a **necessary** condition, and assumes a **differentiable solution**. The sufficiency conditions are quite involved. Again, see the Additional Readings for details. Respect for some of these hazards may be developed by further considering the soap-film problem in Example 22.1.3, with $(x_1, y_1) = (-x_0, 1)$, $(x_2, y_2) = (+x_0, 1)$. We are therefore considering a soap film stretched between two rings of unit radius at $x = \pm x_0$. The problem is to predict the curve $y(x)$ assumed by the soap film.

By referring to Eq. (22.24), we find that $c_2 = 0$ because our problem is symmetric about $x = 0$. Then

$$y = c_1 \cosh \left(\frac{x}{c_1} \right), \quad (22.25)$$

and our endpoint conditions become

$$c_1 \cosh \left(\frac{x_0}{c_1} \right) = 1. \quad (22.26)$$

If we take $x_0 = \frac{1}{2}$ we obtain the following transcendental equation for c_1 :

$$1 = c_1 \cosh \left(\frac{1}{2c_1} \right). \quad (22.27)$$

We find that this equation has two solutions: $c_1 = 0.2350$, leading to a “deep” curve, and $c_1 = 0.8483$, leading to a “shallow” curve. Which curve is assumed by the soap film?

Before answering this question, consider the physical situation with the rings moved apart so that $x_0 = 1$. Then Eq. (22.26) becomes

$$1 = c_1 \cosh\left(\frac{1}{c_1}\right), \quad (22.28)$$

which has **no real solutions**. The physical significance is that as the unit-radius rings were moved out from the origin, a point was reached at which the soap film could no longer maintain the same horizontal force over each vertical section. Stable equilibrium was no longer possible. The soap film broke (irreversible process) and formed a circular film over each ring (with a total area of $2\pi = 6.2832\dots$). This is known as the Goldschmidt discontinuous solution to the soap-film problem.

The next question is: How large may x_0 be and still give a real solution for Eq. (22.26)? Solving Eq. (22.26) for x_0 ,

$$x_0 = c_1 \cosh^{-1}(1/c_1), \quad (22.29)$$

we find that x_0 will be real only for $c_1 \leq 1$ and that its maximum value is attained when $dx_0/dc_1 = 0$. A plot of x_0 vs. c_1 is shown in Fig. 22.5; it helps to explain the behavior we observed at $x_0 = \frac{1}{2}$. We see from the plot (and more precisely from Exercise 22.1.6) that the Euler equation has no solutions for $x_0 > x_{\max}$, where $x_{\max} \approx 0.6627$, and that this x_0 value occurs when $c_1 \approx 0.5524$. For values of x_0 smaller than $x_{\max}$, there are solutions for two different values of c_1 , corresponding to the “deep” and “shallow” curves found earlier for $x_0 = \frac{1}{2}$.

Returning to the question as to which solution of Eq. (22.26) describes the soap film, let us calculate the area corresponding to each solution. Using Eq. (22.22) to reach the last

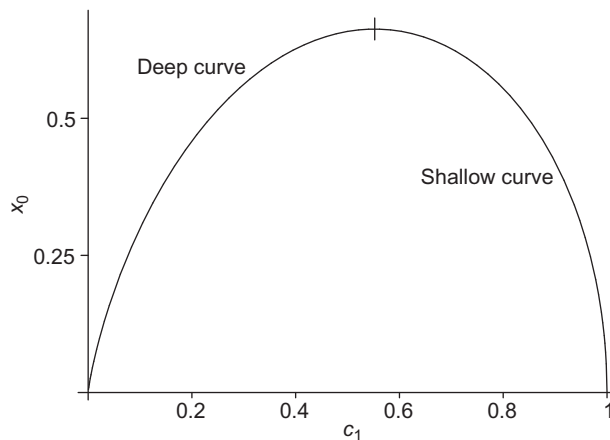


FIGURE 22.5 Solutions of Eq. (22.26) for unit-radius rings at $x = \pm x_0$.

member of the first line below, we have

$$\begin{aligned}
 A &= 4\pi \int_0^{x_0} y(1 + y_x^2)^{1/2} dx = \frac{4\pi}{c_1} \int_0^{x_0} y^2 dx \\
 &= 4\pi c_1 \int_0^{x_0} \left(\cosh \frac{x}{c_1} \right)^2 dx = \pi c_1^2 \left[\sinh \left(\frac{2x_0}{c_1} \right) + \frac{2x_0}{c_1} \right]. \quad (22.30)
 \end{aligned}$$

For $x_0 = \frac{1}{2}$, Eq. (22.30) leads to

$$c_1 = 0.2350 \quad \rightarrow \quad A = 6.8456,$$

$$c_1 = 0.8483 \quad \rightarrow \quad A = 5.9917,$$

showing that the former can at most be only a local minimum. A more detailed investigation (compare Bliss, Additional Readings, chapter IV) shows that this surface is not even a local minimum. For $x_0 = \frac{1}{2}$, the soap film will be described by the shallow curve

$$y = 0.8483 \cosh \left(\frac{x}{0.8483} \right).$$

This shallow catenoid (catenary of revolution) will be an absolute minimum for $0 \leq x_0 < 0.528$. However, for $0.528 < x_0 < 0.6627$, its area is greater than that of the Goldschmidt discontinuous solution (6.2832) and it is only a relative minimum. See Fig. 22.6.

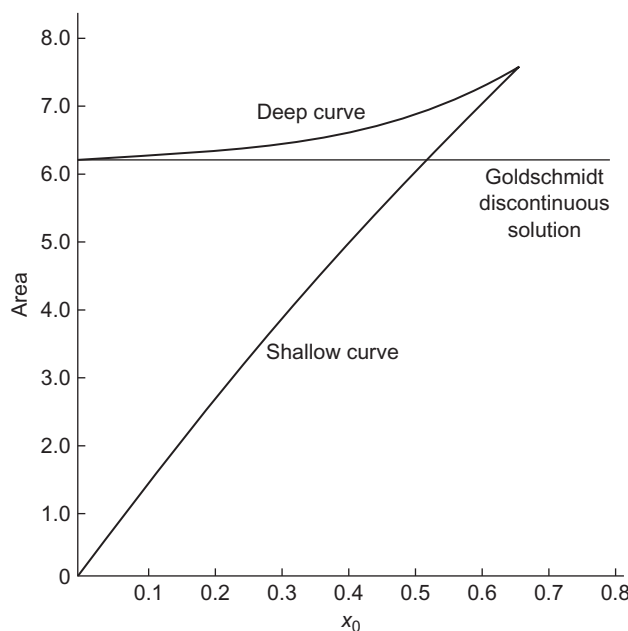


FIGURE 22.6 Catenoid area and that of the discontinuous solution of the soap-film problem (unit-radius rings at $x = \pm x_0$).

For an excellent discussion of both the mathematical problems and experiments with soap films, we refer to Courant and Robbins in Additional Readings. The larger message of this subsection is the extent to which one must use caution in accepting solutions of the Euler equations.

Exercises

- 22.1.1** For $dy/dx \equiv y_x \neq 0$, show the equivalence of the two forms of Euler's equation:

$$\frac{\partial f}{\partial x} - \frac{d}{dx} \frac{\partial f}{\partial y_x} = 0$$

and

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(f - y_x \frac{\partial f}{\partial y_x} \right) = 0.$$

- 22.1.2** Derive Euler's equation by expanding the integrand of

$$J(\alpha) = \int_{x_1}^{x_2} f(y(x, \alpha), y_x(x, \alpha), x) dx$$

in powers of α .

Note. The stationary condition is $\partial J(\alpha)/\partial \alpha = 0$, evaluated at $\alpha = 0$. The terms quadratic in α may be useful in establishing the nature of the stationary solution (maximum, minimum, or saddle point).

- 22.1.3** Find the Euler equation corresponding to Eq. (22.14) if $f = f(y_{xx}, y_x, y, x)$, assuming that y and y_x have fixed values at the endpoints of their interval of definition.

$$\text{ANS. } \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial y_{xx}} \right) - \frac{d}{dx} \left(\frac{\partial f}{\partial y_x} \right) + \frac{\partial f}{\partial y} = 0.$$

- 22.1.4** The integrand $f(y, y_x, x)$ of Eq. (22.2) has the form

$$f(y, y_x, x) = f_1(x, y) + f_2(x, y)y_x.$$

- (a) Show that the Euler equation leads to

$$\frac{\partial f_1}{\partial y} - \frac{\partial f_2}{\partial x} = 0.$$

- (b) What does this imply for the dependence of the integral J on the choice of path?

- 22.1.5** Show that the condition that $J = \int f(x, y) dx$ has a stationary value

- (a) leads to $f(x, y)$ independent of y and
 (b) yields no information about any x -dependence.

We get no (continuous, differentiable) solution. To be a meaningful variational problem, dependence on y or higher derivatives is essential.

Note. The situation will change when constraints are introduced (compare to [Exercise 22.4.6](#)).

- 22.1.6** A soap film stretched between two rings of unit radius centered at $\pm x_0$ will have its closest approach to the x -axis at $x = 0$, with the distance from the axis given by c_1 , with x_0 and c_1 related by [Eq. \(22.26\)](#) or [Eq. \(22.29\)](#).

- (a) Show that dc_1/dx_0 becomes infinite when $x_0 \sinh(x_0/c_1) = 1$, indicating that the soap film becomes unstable if x_0 is increased beyond the value satisfying this condition.
- (b) Show that the condition of part (a) is equivalent to

$$\frac{x_0}{c_1} = \coth\left(\frac{x_0}{c_1}\right).$$

- (c) Solve the transcendental equation of part (b) to obtain the critical value of x_0/c_1 and show that the separate values of x_0 and c_1 are then approximately $x_0 \approx 0.6627$ and $c_1 \approx 0.5524$.

- 22.1.7** A soap film is stretched across the space between two rings of unit radius centered at $\pm x_0$ on the x -axis and perpendicular to the x -axis. Using the solution developed in [Example 22.1.3](#), set up the transcendental equations for the condition that x_0 is such that the area of the curved surface of rotation equals the area of the two rings (Goldschmidt discontinuous solution). Solve for x_0 .

- 22.1.8** In [Example 22.1.1](#), expand $J[y(x, \alpha)] - J[y(x, 0)]$ in powers of α . The term linear in α leads to the Euler equation and to the straight-line solution, [Eq. \(22.16\)](#). Investigate the α^2 term and show that the stationary value of J , the straight-line distance, is a **minimum**.

- 22.1.9** (a) Show that the integral

$$J = \int_{x_1}^{x_2} f(y, y_x, x) dx, \quad \text{with } f = y(x),$$

has **no** extreme values.

- (b) If $f(y, y_x, x) = y^2(x)$, find a discontinuous solution similar to the Goldschmidt solution for the soap-film problem.

- 22.1.10** Fermat's principle of optics states that a light ray in a medium for which n is the (position-dependent) index of refraction will follow the path $y(x)$ for which

$$\int_{x_1, y_1}^{x_2, y_2} n(y, x) ds$$

is a minimum. For $y_2 = y_1 = 1$, $-x_1 = x_2 = 1$, find the ray path if

- (a) $n = e^y$, (b) $n = a(y - y_0)$, $y > y_0$.

- 22.1.11** A particle moves, starting at rest, from point A on the surface of the Earth to point B (also on the surface) by sliding frictionlessly through a tunnel. Find the differential

equation satisfied by the path if the transit time is to be a minimum. Assume the Earth to be a nonrotating sphere of uniform density.

Hint. The potential energy of a particle of mass m a distance $r < R$ from the center of the Earth, with R the Earth's radius, is $\frac{1}{2}mg(R^2 - r^2)/R$, where g is the gravitational acceleration at the Earth's surface. It is convenient to describe the path of the particle (in the plane through A , B , and the center of the Earth) by plane polar coordinates (r, θ) , with A at $(R, -\varphi)$ and B at (R, φ) .

ANS. Letting r_0 be the minimum value of r (reached at $\theta = 0$),

$$\text{Eq. (22.21) yields } r_\theta^2 = \frac{r^2 R^2 (r^2 - r_0^2)}{r_0^2 (R^2 - r^2)} \quad (\text{the constant in}$$

that equation has the value such that $r_\theta = 0$ at $\theta = 0$).

The solution for the path is a hypocycloid, generated by a circle of radius $\frac{1}{2}(R - r_0)$ rolling inside the circle of radius R . You might like to show that the transit time is

$$t = \pi \frac{(R^2 - r_0^2)^{1/2}}{(Rg)^{1/2}}.$$

For details see P. W. Cooper, *Am. J. Phys.* **34**: 68 (1966); G. Veneziano, *et al.*, **34**: 701 (1966).

- 22.1.12** A ray of light follows a straight-line path in a first homogeneous medium, is refracted at an interface, and then follows a new straight-line path in the second medium. See Fig. 22.7. Use Fermat's principle of optics to derive Snell's law of refraction:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2.$$

Hint. Keep the points (x_1, y_1) and (x_2, y_2) fixed and vary x_0 to satisfy Fermat's principle.

Note. This is **not** an Euler equation problem, because the light path is not differentiable at x_0 .

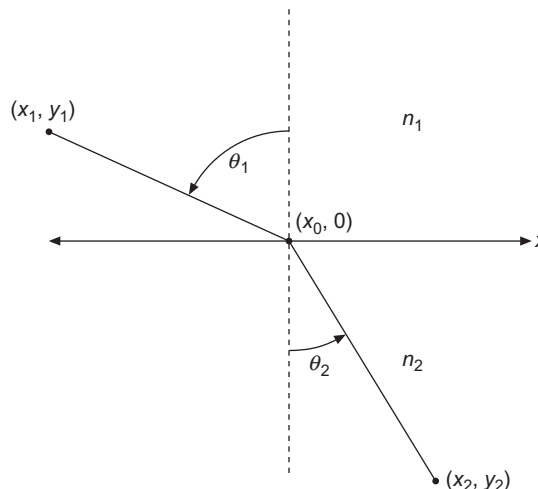


FIGURE 22.7 Snell's law.

- 22.1.13** A second soap-film configuration for the unit-radius rings at $x = \pm x_0$ consists of a circular disk, radius a , in the $x = 0$ plane and two catenoids of revolution, one joining the disk and each ring. One catenoid may be described by

$$y = c_1 \cosh \left(\frac{x}{c_1} + c_3 \right).$$

- (a) Impose boundary conditions at $x = 0$ and $x = x_0$.
- (b) Although not necessary, it is convenient to require that the catenoids form an angle of 120° where they join the central disk. Express this third boundary condition in mathematical terms.
- (c) Show that the total area of catenoids plus central disk is then

$$A = c_1^2 \left[\sinh \left(\frac{2x_0}{c_1} + 2c_3 \right) + \frac{2x_0}{c_1} \right].$$

Note. Although this soap-film configuration is physically realizable and stable, the area is larger than that of the simple catenoid for all ring separations for which both films exist.

$$\text{ANS. (a) } \begin{cases} 1 = c_1 \cosh \left(\frac{x_0}{c_1} + c_3 \right) \\ a = c_1 \cosh c_3 \end{cases} \quad \text{(b) } \frac{dy}{dx} = \tan 30^\circ = \sinh c_3.$$

- 22.1.14** For the soap film described in [Exercise 22.1.13](#), find (numerically) the maximum value of x_0 .

Note. This calls for a calculator with hyperbolic functions or a table of hyperbolic cotangents.

$$\text{ANS. } x_{0\max} = 0.4078.$$

- 22.1.15** Find the curve of quickest descent from $(0, 0)$ to (x_0, y_0) for a particle that, starting from rest, slides under gravity and without friction. Show that the ratio of times taken by the particle along a straight line joining the two points compared to along the curve of quickest descent is $(1 + 4/\pi^2)^{1/2}$.

Hint. Take y to increase downwards. Apply [Eq. \(22.21\)](#) to obtain $y_x^2 = (1 - c^2 y)/c^2 y$, where c is an integration constant. It is helpful to make the substitution $c^2 y = \sin^2 \varphi/2$ and take $(x_0, y_0) = (\pi/2c^2, 1/c^2)$.

22.2 MORE GENERAL VARIATIONS

Several Dependent Variables

To apply variational methods to classical mechanics, we need to generalize the Euler equation to situations in which there is more than one dependent variable in roles like y in

Eq. (22.2). The generalization corresponds to functionals J of the form

$$J = \int_{x_1}^{x_2} f\left(u_1(x), u_2(x), \dots, u_n(x), u_{1x}(x), u_{2x}(x), \dots, u_{nx}(x), x\right) dx. \quad (22.31)$$

We are now calling the dependent variables u_i to be consistent with notations we will shortly introduce, and as before we use the subscript x to denote differentiation with respect to x , so that $u_{ix} = du_i/dx$ and (later) $\eta_{ix} = d\eta_i/dx$. As in Section 22.1, we determine stationary values of J by comparing neighboring paths for each u_i . Let

$$u_i(x, \alpha) = u_i(x, 0) + \alpha \eta_i(x), \quad i = 1, 2, \dots, n, \quad (22.32)$$

with the η_i independent of one other but subject to the continuity and endpoint restrictions discussed in Section 22.1. By differentiating J from Eq. (22.31) with respect to α and setting $\alpha = 0$ (the condition that J be stationary), we obtain

$$\int_{x_1}^{x_2} \sum_i \left(\frac{\partial f}{\partial u_i} \eta_i + \frac{\partial f}{\partial u_{ix}} \eta_{ix} \right) dx = 0. \quad (22.33)$$

Again, each of the terms $(\partial f / \partial u_{ix}) \eta_{ix}$ is integrated by parts. The integrated part vanishes and Eq. (22.33) becomes

$$\int_{x_1}^{x_2} \sum_i \left(\frac{\partial f}{\partial u_i} - \frac{d}{dx} \frac{\partial f}{\partial u_{ix}} \right) \eta_i dx = 0. \quad (22.34)$$

Since the η_i are arbitrary and **independent** of one another,⁶ each of the terms in the sum must vanish **independently**. We have

$$\frac{\partial f}{\partial u_i} - \frac{d}{dx} \frac{\partial f}{\partial u_{ix}} = 0, \quad i = 1, 2, \dots, n, \quad (22.35)$$

a whole set of Euler equations, each of which must be satisfied for a stationary value of J .

Hamilton's Principle

The most important application of Eq. (22.31) occurs when the integrand f is taken to be a Lagrangian L . The Lagrangian (for nonrelativistic systems; see Exercise 22.2.5 for a relativistic particle) is defined as the **difference** of kinetic and potential energies of a system:

$$L \equiv T - V. \quad (22.36)$$

Using time as an independent variable instead of x and $x_i(t)$ as the dependent variables, our conversion of Eq. (22.31) involves the replacements

$$x \rightarrow t, \quad y_i \rightarrow x_i(t), \quad y_{ix} \rightarrow \dot{x}_i(t);$$

⁶For example, we could set $\eta_2 = \eta_3 = \eta_4 \cdots = 0$, eliminating all but one term of the sum, and then treat η_1 exactly as in Section 22.1.

$x_i(t)$ is the position and $\dot{x}_i = dx_i/dt$ is the velocity of particle i as a function of time. The equation $\delta J = 0$ is then a mathematical statement of Hamilton's principle of classical mechanics,

$$\delta \int_{t_1}^{t_2} L(x_1, x_2, \dots, x_n, \dot{x}_1, \dot{x}_2, \dots, \dot{x}_n; t) dt = 0. \quad (22.37)$$

In words, Hamilton's principle asserts that the motion of the system from time t_1 to t_2 is such that the time integral of the Lagrangian L , or action, has a stationary value. The resulting Euler equations are usually called the **Lagrangian equations of motion**,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} - \frac{\partial L}{\partial x_i} = 0 \quad (\text{each } i). \quad (22.38)$$

These Lagrangian equations can be derived from Newton's equations of motion, and Newton's equations can be derived from Lagrange's. The two sets of equations are equally "fundamental."

The Lagrangian formulation has advantages over the conventional Newtonian laws. Whereas Newton's equations are vector equations, we see that Lagrange's equations involve only scalar quantities. The coordinates $x_1, x_2, \dots$ need not be a standard set of coordinates or lengths. They can be selected to match the conditions of the physical problem. The Lagrange equations are invariant with respect to the choice of coordinate system. Newton's equations (in component form) are not manifestly invariant. For example, Exercise 3.10.27 shows what happens when $\mathbf{F} = m\mathbf{a}$ is resolved in spherical polar coordinates.

Exploiting the concept of energy, we may easily extend the Lagrangian formulation from mechanics to diverse fields, such as electrical networks and acoustical systems. Extensions to electromagnetism appear in the exercises. The result is a unification of otherwise separate areas of physics. In the development of new areas, the quantization of Lagrangian particle mechanics provided a model for the quantization of electromagnetic fields and led to the gauge theory of quantum electrodynamics.

One of the most valuable advantages of Hamilton's principle (the Lagrange equation formulation) is the ease in seeing a relation between a symmetry and a conservation law. As an example, let $x_i = \varphi$, an azimuthal angle. If our Lagrangian is independent of φ (that is, φ is said to be an **ignorable coordinate**), there are two consequences: (1) the conservation or invariance of the component of angular momentum associated with (**conjugate to**) φ , and (2) from Eq. (22.38), $\partial L / \partial \dot{\varphi} = \text{constant}$. Similarly, invariance under translation leads to conservation of linear momentum.

Example 22.2.1 MOVING PARTICLE, CARTESIAN COORDINATES

A particle of mass m moves in one dimension with its position described by a Cartesian coordinate x , subject to a potential $V(x)$. Its kinetic energy is given by $T = m\dot{x}^2/2$, so its Lagrangian L has the form

$$L = T - V = \frac{1}{2}m\dot{x}^2 - V(x).$$

We will need

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial L}{\partial x} = -\frac{dV(x)}{dx} = F(x). \quad (22.39)$$

We have identified the force F as the negative gradient of the potential. Inserting the results from Eq. (22.39) into the Lagrangian equation of motion, Eq. (22.38), we get

$$\frac{d}{dt}(m\dot{x}) - F(x) = 0,$$

which is Newton's second law of motion. ■

Example 22.2.2 MOVING PARTICLE, CIRCULAR CYLINDRICAL COORDINATES

Now let us consider a particle of mass m moving in the xy -plane, that is, $z = 0$. We use cylindrical coordinates ρ, φ . The kinetic energy is

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m(\dot{\rho}^2 + \rho^2\dot{\varphi}^2), \quad (22.40)$$

and we take $V = 0$ for simplicity.

We could have converted $\dot{x}^2 + \dot{y}^2$ into circular cylindrical coordinates by taking $x(\rho, \varphi) = \rho \cos \varphi$, $y(\rho, \varphi) = \rho \sin \varphi$, and then differentiating with respect to time and squaring. What we actually did was to recognize that the cylindrical coordinates are an orthogonal system with scale factors $h_\rho = 1$, $h_\varphi = \rho$, so the velocity v has in the cylindrical system components $v_\rho = \dot{\rho}$ and $v_\varphi = \rho\dot{\varphi}$.

We now apply the Lagrangian equations of motion first to the ρ coordinate and then to φ :

$$\frac{d}{dt}(m\dot{\rho}) - m\rho\dot{\varphi}^2 = 0, \quad \frac{d}{dt}(m\rho^2\dot{\varphi}) = 0.$$

The second equation is a statement of conservation of angular momentum. The first may be interpreted as radial acceleration⁷ equated to centrifugal force. In this sense the centrifugal force is a real force. It is of some interest that this interpretation of centrifugal force as a real force is supported by the general theory of relativity. ■

Hamilton's Equations

Hamilton was the first to show that Euler's equation for the Lagrangian enabled the equations of motion to be reduced to the set of coupled first-order PDEs called **Hamilton's equations**. A starting point for this analysis is the definition of the **canonical momentum** p_i conjugate to the coordinate q_i , defined as

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \quad (22.41)$$

⁷Here is a second method of attacking Exercise 3.10.13.

This definition is consistent with the elementary definition of momentum in Cartesian coordinates, where (in one dimension) $T = m\dot{q}^2/2$, $p = m\dot{q}$. From Eq. (22.41) and the Lagrangian equations of motion, Eq. (22.38), we have by direct substitution

$$\dot{p}_i = \frac{\partial L}{\partial q_i}, \quad (22.42)$$

and this permits us to write the variation of L in the form

$$dL = \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \frac{\partial L}{\partial t} dt = \sum_i (\dot{p}_i dq_i + p_i d\dot{q}_i) + \frac{\partial L}{\partial t} dt. \quad (22.43)$$

We now define the **Hamiltonian** as

$$H = \sum_i p_i \dot{q}_i - L, \quad (22.44)$$

and compute

$$dH = \sum_i (p_i d\dot{q}_i + \dot{q}_i dp_i) - \left(\sum_i (\dot{p}_i dq_i + p_i d\dot{q}_i) + \frac{\partial L}{\partial t} dt \right) = \sum_i (\dot{q}_i dp_i - \dot{p}_i dq_i) - \frac{\partial L}{\partial t} dt. \quad (22.45)$$

But from the chain rule for differentiation, we also have

$$dH = \sum_i \left(\frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial q_i} dq_i \right) + \frac{\partial H}{\partial t} dt. \quad (22.46)$$

Equating the coefficients of dp_i , dq_i , and dt in Eqs. (22.45) and (22.46), we obtain Hamilton's equations:

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\dot{p}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (22.47)$$

In conservative systems, $\partial H/\partial t = 0$, and H has a constant value equal to the total energy of the system.

Several Independent Variables

Sometimes the integrand f in an equation analogous to Eq. (22.2) will contain an unknown function, u , that is a function of several independent variables, $u = u(x, y, z)$. In the three-dimensional case, for example, that equation becomes

$$J = \iiint f(u, u_x, u_y, u_z, x, y, z) dx dy dz, \quad (22.48)$$

where $u_x = \partial u/\partial x$, $u_y = \partial u/\partial y$, $u_z = \partial u/\partial z$, and u is assumed to have specified values on the boundary of the region of integration.

Generalizing the analysis of Section 22.1, we represent the variation of u as

$$u(x, y, z, \alpha) = u(x, y, z, 0) + \alpha \eta(x, y, z),$$

where η is arbitrary except that it must vanish on the boundary. Our integral J is now, as in Section 22.1, a function of α , and our variational problem is to make J stationary with respect to α .

Differentiating the integral Eq. (22.48) with respect to the parameter α and then setting $\alpha = 0$, we obtain

$$\left. \frac{\partial J}{\partial \alpha} \right|_{\alpha=0} = \iiint \left(\frac{\partial f}{\partial u} \eta + \frac{\partial f}{\partial u_x} \eta_x + \frac{\partial f}{\partial u_y} \eta_y + \frac{\partial f}{\partial u_z} \eta_z \right) dx dy dz = 0.$$

We continue to use a notation similar to that used previously: η_x is shorthand for $\partial\eta/\partial x$, etc.

Again, we integrate each of the terms $(\partial f/\partial u_i) \eta_i$ by parts. The integrated part vanishes at the boundary (because the deviation η is required to go to zero there) and we get

$$\iiint \left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial x} \frac{\partial f}{\partial u_x} - \frac{\partial}{\partial y} \frac{\partial f}{\partial u_y} - \frac{\partial}{\partial z} \frac{\partial f}{\partial u_z} \right) \eta(x, y, z) dx dy dz = 0. \quad (22.49)$$

We must now digress to clarify the notation in Eq. (22.49). The derivative $\partial/\partial x$ enters that equation as a result of the integration by parts, and it therefore must act on all the x dependence of $\partial f/\partial u_x$, not just on the explicit appearance of x in f . The reader may recall that this derivative was written d/dx when it arose in Section 22.1, but that notation is not entirely appropriate here as the functions involved also depend on y and z .

We conclude our analysis with the now-familiar observation that since the variation $\eta(x, y, z)$ is arbitrary, the term in large parentheses is set equal to zero. This yields the Euler equation for (three) independent variables,

$$\frac{\partial f}{\partial u} - \frac{\partial}{\partial x} \frac{\partial f}{\partial u_x} - \frac{\partial}{\partial y} \frac{\partial f}{\partial u_y} - \frac{\partial}{\partial z} \frac{\partial f}{\partial u_z} = 0. \quad (22.50)$$

Remember that the derivative $\partial/\partial x$ operates on both the explicit and implicit x dependence of $\partial f/\partial u_x$; similar remarks apply to $\partial/\partial y$ and $\partial/\partial z$.

Example 22.2.3 LAPLACE'S EQUATION

A variational problem with several independent variables is provided by electrostatics. An electrostatic field has

$$\text{energy density} = \frac{1}{2} \epsilon \mathbf{E}^2,$$

where $\mathbf{E}$ is the electric field. In terms of the static potential φ ,

$$\text{energy density} = \frac{1}{2} \epsilon (\nabla \varphi)^2.$$

Now let us impose the requirement that the electrostatic energy (associated with the field) in a given charge-free volume be a minimum subject to specific conditions on φ at the boundary. The assumption that the volume is charge-free makes φ continuous and differentiable throughout the volume, and we therefore have a situation to which an Euler equation applies. We have the volume integral

$$J = \iiint (\nabla \varphi)^2 dx dy dz = \iiint (\varphi_x^2 + \varphi_y^2 + \varphi_z^2) dx dy dz,$$

where φ_x stands for $\partial\varphi/\partial x$. Thus,

$$f(\varphi, \varphi_x, \varphi_y, \varphi_z, x, y, z) = \varphi_x^2 + \varphi_y^2 + \varphi_z^2,$$

so Euler's equation, Eq. (22.50), yields (with u in that equation replaced by φ)

$$-2(\varphi_{xx} + \varphi_{yy} + \varphi_{zz}) = 0,$$

which in the usual vector notation is equivalent to

$$\nabla^2 \varphi(x, y, z) = 0.$$

This is Laplace's equation of electrostatics.

Closer investigation shows that this stationary value is indeed a minimum. Thus the demand that the field energy be minimized leads to Laplace's PDE. ■

Several Dependent and Independent Variables

In some cases our integrand f contains more than one dependent variable and more than one independent variable. Consider

$$f = f\left(p(x, y, z), p_x, p_y, p_z, q(x, y, z), q_x, q_y, q_z, r(x, y, z), r_x, r_y, r_z, x, y, z\right). \quad (22.51)$$

We proceed as before with

$$p(x, y, z, \alpha) = p(x, y, z, 0) + \alpha\xi(x, y, z),$$

$$q(x, y, z, \alpha) = q(x, y, z, 0) + \alpha\eta(x, y, z),$$

$$r(x, y, z, \alpha) = r(x, y, z, 0) + \alpha\zeta(x, y, z), \quad \text{and so on.}$$

Keeping in mind that ξ, η , and ζ are independent of one another, as were the η_i in Eq. (22.32), the same differentiation and then integration by parts will lead to

$$\frac{\partial f}{\partial p} - \frac{\partial}{\partial x} \frac{\partial f}{\partial p_x} - \frac{\partial}{\partial y} \frac{\partial f}{\partial p_y} - \frac{\partial}{\partial z} \frac{\partial f}{\partial p_z} = 0, \quad (22.52)$$

with similar equations for functions q and r . Replacing $p, q, r, \dots$ with y_i and $x, y, z, \dots$ with x_i , we can put Eq. (22.52) in a more compact form:

$$\frac{\partial f}{\partial y_i} - \sum_j \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial y_{ij}} \right) = 0, \quad i = 1, 2, \dots, \quad (22.53)$$

in which

$$y_{ij} \equiv \frac{\partial y_i}{\partial x_j}.$$

An application of Eq. (22.53) appears in Exercise 22.2.10.

Geodesics

Particularly in general relativity, it is of interest to identify the shortest path between two points in a “curved space,” i.e., a space characterized by a metric tensor more general than that of Euclidean or even Minkowski space. A path that is a “local minimum” (calculated using the relevant metric), meaning that it is shorter than other paths that can be reached from it by small deformations, is referred to as a **geodesic**. This definition causes both the two great-circle paths of Fig. 22.2 to be identified as geodesics, because even the longer path is of minimum length relative to small deformations. In practice, it is usually easy to identify which of several geodesics in fact corresponds to the shortest path.

The calculus of variations is the natural tool for identifying geodesics, and in fact it was used in Example 22.1.1 to verify that a straight line is the geodesic connecting given points in Euclidean space. To extend the analysis to more general metric spaces, we start by relating the distance between two neighboring points, ds , with the changes in their coordinates, dq^i , $i = 1, 2, \dots$. Note that we distinguish between covariant and contravariant quantities, using superscripts for the latter (coordinate displacements are contravariant; compare with Section 4.3). The distance ds is a scalar, given by

$$ds^2 = g_{ij} dq^i dq^j. \quad (22.54)$$

Here g_{ij} is the metric tensor, which is symmetric but in many cases of interest not diagonal. Note that we are using the Einstein summation convention, so i and j in Eq. (22.54) are summed, causing ds^2 to be a scalar. This formula is an obvious generalization of that for Euclidean space,

$$ds^2 = dx^2 + dy^2 + dz^2,$$

but differs therefrom in that the coordinates q_i are not assumed to be mutually orthogonal, so ds^2 contains cross terms $dq^i dq^j$ with $i \neq j$.

A path in our curved space can be described parametrically by giving the q_i as functions of an independent variable that we will call u , and the distance between two points A and B can then be represented as

$$\begin{aligned} J &= \int_A^B \frac{ds}{du} du = \int_A^B \frac{\sqrt{g_{ij} dq^i dq^j}}{du} du = \int_A^B \sqrt{g_{ij} \frac{dq^i}{du} \frac{dq^j}{du}} du \\ &= \int_A^B \sqrt{g_{ij} \dot{q}^i \dot{q}^j} du, \end{aligned} \quad (22.55)$$

where we are borrowing the dot notation, $\dot{q}^i \equiv dq^i/du$.

One could now proceed to find the $q^i(u)$ that minimize J , but this is a relatively difficult problem. Instead we rely on the Lagrangian formulation of relativistic mechanics, where, for a particle not subject to a potential (other than a gravitational force whose effect is described by the metric), the Lagrangian reduces to

$$L = \frac{m}{2} g_{ij} \dot{q}^i \dot{q}^j. \quad (22.56)$$

Here the dot notation refers to derivatives with respect to the proper time τ (or to any other variable related thereto by an **affine transformation** (meaning the new variable, e.g., u , is related to τ by a transformation of the form $u = a\tau + b$). This means that we can replace the minimization of J by that of the **action**:

$$\delta \int_A^B g_{ij} \dot{q}^i \dot{q}^j du = 0, \quad (22.57)$$

in effect simplifying our problem by eliminating the radical that was present in Eq. (22.55).

The minimization in Eq. (22.57) is a relatively simple standard problem in the calculus of variations; for solving it we note that each g_{ij} is in general a function of all the q^k (but not the derivatives $\dot{q}^k$). There will be an Euler equation for each k ; before simplification they take the form

$$\frac{\partial g_{ij} \dot{q}^i \dot{q}^j}{\partial q^k} - \frac{d}{du} \frac{\partial g_{ij} \dot{q}^i \dot{q}^j}{\partial \dot{q}^k} = 0. \quad (22.58)$$

Starting to evaluate Eq. (22.58), we get

$$\frac{\partial g_{ij}}{\partial q^k} \dot{q}^i \dot{q}^j - \frac{d}{du} g_{ij} \frac{\partial}{\partial \dot{q}^k} (\dot{q}^i \dot{q}^j) = \frac{\partial g_{ij}}{\partial q^k} \dot{q}^i \dot{q}^j - \frac{d}{du} (g_{kj} \dot{q}^j + g_{ik} \dot{q}^i) = 0. \quad (22.59)$$

Some simplification is achieved by using the relations

$$\frac{d\dot{q}^j}{du} = \ddot{q}^j \quad \text{and} \quad \frac{dg_{kj}}{du} = \frac{\partial g_{kj}}{\partial q^i} \dot{q}^i$$

(remember that the Einstein summation convention is still in use). Equation (22.59) reduces to

$$\frac{1}{2} \dot{q}^i \dot{q}^j \left[\frac{\partial g_{ij}}{\partial q^k} - \frac{\partial g_{kj}}{\partial q^i} - \frac{\partial g_{ik}}{\partial q^j} \right] - g_{ik} \ddot{q}^i = 0. \quad (22.60)$$

As a final simplification, we multiply Eq. (22.60) by g^{kl} and use the identity $g^{kl} g_{ik} = \delta_i^l$, reaching (in a more expanded notation) the **geodesic equation**

$$\frac{d^2 q^l}{du^2} + \frac{dq^i}{du} \frac{dq^j}{du} \frac{1}{2} g^{kl} \left[\frac{\partial g_{kj}}{\partial q^i} + \frac{\partial g_{ik}}{\partial q^j} - \frac{\partial g_{ij}}{\partial q^k} \right] = 0. \quad (22.61)$$

Comparing with the formula for the Christoffel symbol, Eq. (4.63), we can rewrite Eq. (22.61) as

$$\frac{d^2 q^l}{du^2} + \frac{dq^i}{du} \frac{dq^j}{du} \Gamma_{ij}^l = 0. \quad (22.62)$$

Note that although Eq. (22.62) gives the differential equation describing geodesics in curved space, it is a long way from that equation to its explicit solution for significant problems in general relativity. The exploration of such solutions is a topic of current research and beyond the scope of the present text.

Relation to Physics

The calculus of variations as developed so far provides an elegant description of a wide variety of physical phenomena. The physics includes classical mechanics, as in [Examples 22.2.1](#) and [22.2.2](#); relativistic mechanics, [Exercise 22.2.5](#); electrostatics, [Example 22.2.3](#); and electromagnetic theory in [Exercise 22.2.10](#). The convenience should not be minimized, but at the same time we should be aware that in these cases the calculus of variations has only provided an alternate description of what was already known. The situation does change with incomplete theories.

If the basic physics is not yet known, a postulated variational principle can be a useful starting point.

Exercises

- 22.2.1** (a) Develop the equations of motion corresponding to $L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$.
- (b) In what sense do your solutions minimize the integral $\int_{t_1}^{t_2} L \, dt$? Compare the result for your solution with $x = \text{constant}$, $y = \text{constant}$.
- 22.2.2** From the Lagrangian equations of motion, [Eq. \(22.38\)](#), show that a system in stable equilibrium has a minimum potential energy.
- 22.2.3** Write out the Lagrangian equations of motion of a particle in spherical coordinates for potential V equal to a constant. Identify the terms corresponding to (a) centrifugal force and (b) Coriolis force.
- 22.2.4** The spherical pendulum consists of a mass on a wire of length l , free to move in polar angle θ and azimuth angle φ ([Fig. 22.8](#)).
- (a) Set up the Lagrangian for this physical system.
- (b) Develop the Lagrangian equations of motion.

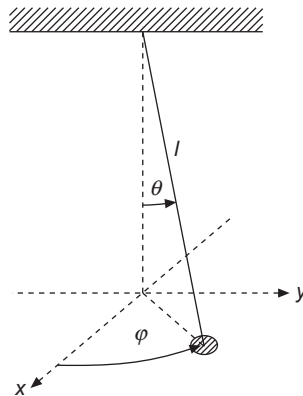


FIGURE 22.8 Spherical pendulum.

22.2.5 Show that the Lagrangian

$$L = m_0 c^2 \left(1 - \sqrt{1 - \frac{v^2}{c^2}} \right) - V(\mathbf{r})$$

leads to a relativistic form of Newton's second law of motion,

$$\frac{d}{dt} \left(\frac{m_0 v_i}{\sqrt{1 - v^2/c^2}} \right) = F_i,$$

in which the force components are $F_i = -\partial V / \partial x_i$.

22.2.6 The Lagrangian for a particle with charge q in an electromagnetic field described by scalar potential φ and vector potential $\mathbf{A}$ is

$$L = \frac{1}{2} m v^2 - q\varphi + q\mathbf{A} \cdot \mathbf{v}.$$

Find the equation of motion of the charged particle.

Hint. $(d/dt)A_j = \partial A_j / \partial t + \sum_i (\partial A_j / \partial x_i) \dot{x}_i$. The dependence of the force fields $\mathbf{E}$ and $\mathbf{B}$ on the potentials φ and $\mathbf{A}$ is developed in Section 3.9; see in particular Eq. (3.108).

$$\text{ANS. } m\ddot{x}_i = q[\mathbf{E} + \mathbf{v} \times \mathbf{B}]_i.$$

22.2.7 Consider a system in which the Lagrangian is given by

$$L(q_i, \dot{q}_i) = T(q_i, \dot{q}_i) - V(q_i),$$

where q_i and $\dot{q}_i$ represent sets of variables. The potential energy V is independent of velocity and neither T nor V has any explicit time dependence.

(a) Show that

$$\frac{d}{dt} \left(\sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L \right) = 0.$$

(b) The constant quantity

$$\sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L$$

defines the Hamiltonian H . Show that under the preceding assumed conditions, H satisfies $H = T + V$, and is therefore the total energy.

Note. The kinetic energy T is a quadratic function of the $\dot{q}_i$.

22.2.8 The Lagrangian for a vibrating string (small-amplitude vibrations) is

$$L = \int \left(\frac{1}{2} \rho u_t^2 - \frac{1}{2} \tau u_x^2 \right) dx,$$

where ρ is the (constant) linear mass density and τ is the (constant) tension. The x -integration is over the length of the string. Show that application of Hamilton's

principle to the Lagrangian density (the integrand), now with two independent variables, leads to the classical wave equation,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\rho}{\tau} \frac{\partial^2 u}{\partial t^2}.$$

- 22.2.9** Show that the stationary value of the total energy of the electrostatic field of [Example 22.2.3](#) is a **minimum**.

Hint. Investigate the α^2 terms of J .

- 22.2.10** The Lagrangian (per unit volume) of an electromagnetic field with a charge density ρ and current density $\mathbf{J}$ is given by

$$L = \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 - \frac{1}{\mu_0} \mathbf{B}^2 \right) - \rho \varphi + \mathbf{J} \cdot \mathbf{A}.$$

Show that Lagrange's equations lead to two of Maxwell's equations. (The remaining two are a consequence of the definition of $\mathbf{E}$ and $\mathbf{B}$ in terms of $\mathbf{A}$ and φ .)

Hint. Take φ and the components of $\mathbf{A}$ as **dependent** variables; and x , y , z , and t as **independent** variables. $\mathbf{E}$ and $\mathbf{B}$ are given in terms of $\mathbf{A}$ and φ by Eq. (3.108).

22.3 CONSTRAINED MINIMA/MAXIMA

In preparation for dealing with problems in the calculus of variations in which an integral is to be minimized subject to constraints (which may either be algebraic equations or fixed values of other integrals), we look now at situations in which we seek a constrained extremum of an ordinary function.

A typical constrained problem of the type now under consideration is the minimization of a function of several variables, here illustrated as $f(x, y, z)$, subject to the constraint that $g(x, y, z)$ be kept constant. Since the equation $g(x, y, z) = C$ defines a surface, our constrained problem is that of minimizing $f(x, y, z)$ on a surface of constant g . The presence of the constraint means that only two of the three variables x , y , z are actually independent, and in principle one could solve the constraint equation to obtain z as a function of x and y : $z = z(x, y)$, after which one could obtain the desired minimum by setting to zero the derivatives

$$\frac{\partial}{\partial x} f(x, y, z(x, y)) \quad \text{and} \quad \frac{\partial}{\partial y} f(x, y, z(x, y)).$$

However, it may be cumbersome, or in some cases nearly impossible to solve the constraint equation, and in any case this approach does not treat the variables x , y , z on an explicitly equivalent basis. For these reasons it is useful to employ an alternate procedure, known as the method of **Lagrangian multipliers**.

Lagrangian Multipliers

Continuing with our three-dimensional illustration in which we seek to minimize $f(x, y, z)$ subject to the constraint $g(x, y, z) = C$, our starting point is that the constraint equation implies

$$dg = \left(\frac{\partial g}{\partial x} \right)_{yz} dx + \left(\frac{\partial g}{\partial y} \right)_{xz} dy + \left(\frac{\partial g}{\partial z} \right)_{xy} dz = 0,$$

where (as indicated here explicitly) the partial derivatives of g are taken viewing x , y , and z as independent. Proceeding as for the derivation of Eq. (1.144), we have

$$\left(\frac{\partial z}{\partial x} \right)_y = - \frac{\left(\frac{\partial g}{\partial x} \right)_{yz}}{\left(\frac{\partial g}{\partial z} \right)_{xy}} \quad \text{and} \quad \left(\frac{\partial z}{\partial y} \right)_x = - \frac{\left(\frac{\partial g}{\partial y} \right)_{xz}}{\left(\frac{\partial g}{\partial z} \right)_{xy}}. \quad (22.63)$$

Now setting $(\partial f / \partial x)_y$ to zero, we have (imposing the constraint $dg = 0$)

$$\begin{aligned} \left(\frac{\partial f}{\partial x} \right)_y &= \left(\frac{\partial f}{\partial x} \right)_{yz} + \left(\frac{\partial f}{\partial z} \right)_{xy} \left(\frac{\partial z}{\partial x} \right)_y = \left(\frac{\partial f}{\partial x} \right)_{yz} - \frac{\left(\frac{\partial f}{\partial z} \right)_{xy}}{\left(\frac{\partial g}{\partial z} \right)_{xy}} \left(\frac{\partial g}{\partial x} \right)_{yz} \\ &= \left(\frac{\partial f}{\partial x} \right)_{yz} - \lambda \left(\frac{\partial g}{\partial x} \right)_{yz} = 0, \end{aligned} \quad (22.64)$$

where

$$\lambda = \frac{\left(\frac{\partial f}{\partial z} \right)_{xy}}{\left(\frac{\partial g}{\partial z} \right)_{xy}}. \quad (22.65)$$

The quantity λ is called a **Lagrangian multiplier**.

Now taking Eq. (22.64), its equivalent with y replacing x , and a rearranged form of Eq. (22.65), we have the symmetrical set of formulas

$$\begin{aligned} \left(\frac{\partial f}{\partial x} \right)_{yz} - \lambda \left(\frac{\partial g}{\partial x} \right)_{yz} &= 0, \\ \left(\frac{\partial f}{\partial y} \right)_{xz} - \lambda \left(\frac{\partial g}{\partial y} \right)_{xz} &= 0, \\ \left(\frac{\partial f}{\partial z} \right)_{xy} - \lambda \left(\frac{\partial g}{\partial z} \right)_{xy} &= 0. \end{aligned} \quad (22.66)$$

The generalization of Eqs. (22.66) to n variables and k constraints is

$$\frac{\partial f}{\partial x_i} - \sum_{j=1}^k \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad i = 1, 2, \dots, n. \quad (22.67)$$

The n equations, Eqs. (22.67), contain $n + k$ unknowns (the n x_i and the k λ_j), and they are to be solved subject also to the k constraint equations. In some problems it is never necessary to evaluate explicitly the Lagrangian multipliers, and for this reason the method is sometimes referred to as that of **(Lagrange's) undetermined multiplier(s)**.

Note that the formulation provided above does not only identify minima; the same equations will locate maxima and saddle points. It is necessary to determine the nature of the stationary points from the specific problem at hand.

While the derivation of Eq. (22.66) was asymmetric in that λ was obtained considering z to be a dependent variable, we could have carried out the analysis with x or y in place of z . This gives us an alternate route to the final formulas in the special case that $(\partial g/\partial z)$ vanishes, in which case Eq. (22.65) becomes undefined. The method only fails if all the derivatives of a constraint function vanish at the stationary point.

Example 22.3.1 MINIMIZING SURFACE-TO-VOLUME RATIO

Consider a right circular cylinder of radius r and height h . We wish to find the ratio h/r that will minimize the surface area for a fixed enclosed volume. The relevant formulas are: surface area $S = 2\pi(rh + r^2)$, volume $V = \pi r^2 h$.

Applying Eqs. (22.67) for the case of one constraint and two independent variables, we have

$$\frac{\partial S}{\partial r} - \lambda \frac{\partial V}{\partial r} = 2\pi(h + 2r) - \lambda(2\pi r h) = 0,$$

$$\frac{\partial S}{\partial h} - \lambda \frac{\partial V}{\partial h} = 2\pi r - \lambda\pi r^2 = 0.$$

Eliminating λ from these equations, we find $h/r = 2$. Because we have not also used the constraint equation, we get only the ratio of the two variables h and r (which is the information that is relevant for the present problem). However, if we specify the volume V (i.e., use the constraint equation), we then get individual values of h and r .

We close with two more observations: (1) Our solution obviously provides a *minimum* S/V ratio, but in principle this has to be determined by closer study of the problem. In the present case, there is no maximum, as S/V increases without limit as h/r approaches zero. (2) We note that minimizing S for fixed V is the same thing as maximizing V for fixed S , and leads to equivalent Lagrangian multiplier equations. ■

Exercises

22.3.0 The following problems are to be solved by using Lagrangian multipliers.

22.3.1 The ground-state energy of a quantum particle of mass m in a pillbox (right-circular cylinder) is given by

$$E = \frac{\hbar^2}{2m} \left(\frac{(2.4048)^2}{R^2} + \frac{\pi^2}{H^2} \right),$$

in which R is the radius and H is the height of the pillbox. Find the ratio of R to H that will minimize the energy for a fixed volume.

22.3.2 The U.S. Post Office limits first-class mail to Canada to a total of 36 inches, length plus girth. Using Lagrange multipliers, find the dimensions of the rectangular parallelepiped of maximum volume subject to this constraint.

22.3.3 A thermal nuclear reactor is subject to the constraint

$$\varphi(a, b, c) = \left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2 + \left(\frac{\pi}{c}\right)^2 = B^2, \quad \text{a constant,}$$

where the reactor is a rectangular parallelepiped of sides a , b , and c . Find the ratios of a , b , and c that maximize the reactor volume.

ANS. $a = b = c$, cube.

22.3.4 For a lens of focal length f , the object distance p and the image distance q are related by $1/p + 1/q = 1/f$. Find the minimum object-image distance ($p + q$) for fixed f . Assume real object and image (p and q both positive).

22.3.5 You have an ellipse $(x/a)^2 + (y/b)^2 = 1$. Find the inscribed rectangle of maximum area. Show that the ratio of the area of the maximum-area rectangle to the area of the ellipse is $2/\pi = 0.6366$.

22.3.6 A rectangular parallelepiped is inscribed in an ellipsoid of semiaxes a , b , and c . Maximize the volume of the inscribed rectangular parallelepiped. Show that the ratio of the maximum volume to the volume of the ellipsoid is $2/\pi\sqrt{3} \approx 0.367$.

22.3.7 Find the maximum value of the directional derivative of $\varphi(x, y, z)$,

$$\frac{d\varphi}{ds} = \frac{\partial\varphi}{\partial x} \cos \alpha + \frac{\partial\varphi}{\partial y} \cos \beta + \frac{\partial\varphi}{\partial z} \cos \gamma,$$

subject to the constraint,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

22.4 VARIATION WITH CONSTRAINTS

As in earlier sections, we seek the path that will make the integral

$$J = \int f\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) dx_j \quad (22.68)$$

stationary. This is the general case in which x_j represents a set of independent variables and y_i a set of dependent variables. Now, however, we introduce one or more constraints. This means that the y_i are no longer independent of each other. Then, if we vary the y_i by writing $y_i(\alpha) = y_i(0) + \alpha\eta_i$, not all the η_i may then be varied arbitrarily, and the Euler equations would not apply.

Our approach will be to use Lagrange's method of undetermined multipliers. We consider first the possibility that the k th constraint takes the form of an equation:

$$\varphi_k\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) = 0. \quad (22.69)$$

This will ordinarily not be meaningful unless there is more than one dependent or independent variable, so that Eq. (22.69) restricts, but does not fully determine y_i . Remember that y_i and x_j are here used to denote **sets** of variables. To introduce an undetermined multiplier and remain in harmony with our study of the calculus of variations, we note that the constraint, Eq. (22.69), can be stated in the form

$$\int \lambda_k(x_j) \varphi_k\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) dx_j = 0, \quad (22.70)$$

with $\lambda_k(x_j)$ an arbitrary function of the x_j . Equation (22.70) is clearly satisfied if

$$\delta \int \lambda_k(x_j) \varphi_k\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) dx_j = 0. \quad (22.71)$$

Alternatively, we may have a constraint in the form of an integral (now dependent on both the y_i and their derivatives throughout the interval on which the problem is defined):

$$\int \varphi_k\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) dx_j = \text{constant}. \quad (22.72)$$

The effect of this constraint can be brought to a form consistent with Eq. (22.71) by writing

$$\delta \int \lambda_k \varphi_k\left(y_i, \frac{\partial y_i}{\partial x_j}, x_j\right) dx_j = 0. \quad (22.73)$$

Note that in this equation λ_k does not depend on the x_j but is simply a constant, as it is only the integral of φ_k that is required to be stationary.

At this point, our constraints have been written as integrals that are dependent on the undetermined multipliers λ_k , where λ_k means either $\lambda_k(x_j)$ or just λ_k , depending on whether the constraint was from Eq. (22.71) or (22.73). We therefore have our problem in a form suitable for applying the method of Lagrangian multipliers as developed in

Section 22.3, and may use a formula analogous to Eq. (22.67). In our present notation, we obtain

$$\delta \int \left[f \left(y_i, \frac{\partial y_i}{\partial x_j}, x_j \right) + \sum_k \lambda_k \varphi_k \left(y_i, \frac{\partial y_i}{\partial x_j}, x_j \right) \right] dx_j = 0. \quad (22.74)$$

Remember that the Lagrangian multiplier λ_k may depend on the x_j when $\varphi(y_i, x_j)$ is given in the form of Eq. (22.69).

We now continue by treating the entire integrand as a new function whose integral is to be made stationary:

$$g \left(y_i, \frac{\partial y_i}{\partial x_j}, x_j \right) = f + \sum_k \lambda_k \varphi_k. \quad (22.75)$$

If we have N dependent variables y_i ($i = 1, 2, \dots, N$) and m constraints ($k = 1, 2, \dots, m$), then $N - m$ of the η_i may be taken as arbitrary. In place of arbitrary variation of the m remaining η_i , we may instead set the m multipliers λ_k to the (presently unknown) values that permit the Euler equations to be satisfied. The overall result is that we may require satisfaction of an Euler equation for each of the dependent variables y_i , but the m quantities λ_k that appear in the solution of the Euler equations must be assigned values consistent with the constraints that have been imposed. In other words, it will be necessary to solve simultaneously the Euler equations and the equations of constraint to find the function g (and hence f) yielding a stationary value.

Lagrangian Formulation with Constraints

In the absence of constraints, Lagrange's equations of motion Eq. (17.52) were found to be⁸

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0,$$

with t (time) the one independent variable and $q_i(t)$ (the particle positions) a set of dependent variables. Usually the generalized coordinates q_i are chosen to eliminate the forces of constraint, but this is not necessary and not always desirable. In the presence of **holonomic** constraints (those that can be expressed via mathematical expressions, e.g., $\varphi_k = 0$), Hamilton's principle is

$$\delta \int \left[L(q_i, \dot{q}_i, t) + \sum_k \lambda_k(t) \varphi_k(q_i, t) \right] dt = 0, \quad (22.76)$$

and the constrained Lagrangian equations of motion are

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \sum_k a_{ik} \lambda_k. \quad (22.77)$$

⁸The symbol q is customary in classical mechanics. It serves to emphasize that the variable is not necessarily a Cartesian variable (and not necessarily a length).

Usually the constraint is of the form $\varphi_k = \varphi_k(q_i, t)$, independent of the generalized velocities $\dot{q}_i$. In this case the coefficient a_{ik} is given by

$$a_{ik} = \frac{\partial \varphi_k}{\partial q_i}. \quad (22.78)$$

Then $a_{ik}\lambda_k$ (no summation) represents the force of the k th constraint in the $\hat{q}_i$ -direction, appearing in Eq. (22.77) in exactly the same way as $-\partial V/\partial q_i$.

Example 22.4.1 SIMPLE PENDULUM

To illustrate, consider the simple pendulum, a mass m , constrained by a wire of length l to swing in an arc (Fig. 22.9) under a gravitational force characterized by a constant acceleration g . In the absence of the one constraint,

$$\varphi_1 = r - l = 0, \quad (22.79)$$

there are two generalized coordinates r and θ (assuming the motion to be restricted to a vertical plane). The Lagrangian is

$$L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mgr \cos \theta, \quad (22.80)$$

taking the potential V to be zero when the pendulum is horizontal, at $\theta = \pi/2$. Noting that

$$a_{r1} = \frac{\partial \varphi_1}{\partial r} = 1, \quad a_{\theta 1} = \frac{\partial \varphi_1}{\partial \theta} = 0,$$

the equations of motion obtained from Eq. (22.77) are

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = \lambda_1, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0, \quad (22.81)$$

or

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 - mg \cos \theta = \lambda_1,$$

$$\frac{d}{dt}(mr^2\dot{\theta}) + mgr \sin \theta = 0.$$

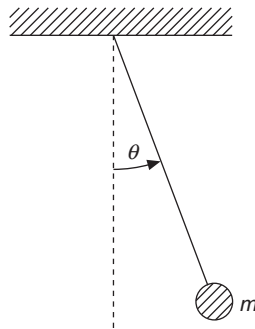


FIGURE 22.9 Simple pendulum.

Substituting from the equation of constraint ($r = l$, $\dot{r} = 0$), these equations become

$$ml\dot{\theta}^2 + mg \cos \theta = -\lambda_1, \quad ml^2\ddot{\theta} + mgl \sin \theta = 0. \quad (22.82)$$

The second equation may be solved for $\theta(t)$ to yield simple harmonic motion if the amplitude is small ($\sin \theta \approx \theta$), whereas the first equation expresses the tension in the wire in terms of θ and $\dot{\theta}$. Note that since the equation of constraint, Eq. (22.79), is in the form of Eq. (22.69), the Lagrange multiplier λ_1 will be a function of t . Since the second equation suffices to determine $\theta(t)$ (assuming a choice of initial conditions), the left-hand side of the first equation can be evaluated if an explicit form for λ_1 is desired. ■

Example 22.4.2 SLIDING OFF A LOG

Another example from mechanics is the problem of a particle sliding on a cylindrical surface, as shown in Fig. 22.10. The object is to find the critical angle θ_c at which the particle flies off from the surface. This critical angle is the angle at which the radial force of constraint goes to zero, and it will depend on the initial velocity with which the particle departs from a position atop the cylinder. To make the problem well-defined, we seek the maximum value that can be attained by θ_c , corresponding to its limit at low initial velocity.

To illustrate the present constrained-minimization method, we take

$$L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - mgr \cos \theta \quad (22.83)$$

and the one equation of constraint,

$$\varphi_1 = r - l = 0. \quad (22.84)$$

Proceeding as in Example 22.4.1, with

$$a_{r1} = \frac{\partial \varphi_1}{\partial r} = 1, \quad a_{\theta 1} = \frac{\partial \varphi_1}{\partial \theta} = 0,$$

we reach

$$m\ddot{r} - mr\dot{\theta}^2 + mg \cos \theta = \lambda_1(\theta),$$

$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} - mgr \sin \theta = 0.$$

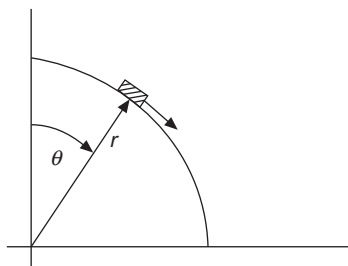


FIGURE 22.10 A particle sliding on a cylindrical surface.

We have chosen to identify the constraining force λ_1 as a function of the angle θ , a valid choice since θ is a single-valued function of the independent variable t .

Inserting the constrained values $r = l$, $\ddot{r} = \dot{r} = 0$, these equations reduce to

$$-ml\dot{\theta}^2 + mg \cos \theta = \lambda_1(\theta), \quad (22.85)$$

$$ml^2\ddot{\theta} - mgl \sin \theta = 0. \quad (22.86)$$

Differentiating Eq. (22.85) with respect to time and remembering that

$$\frac{df(\theta)}{dt} = \frac{df(\theta)}{d\theta} \dot{\theta},$$

we obtain

$$-2ml\ddot{\theta} - mg \sin \theta = \frac{d\lambda_1(\theta)}{d\theta}. \quad (22.87)$$

Combining Eqs. (22.86) and (22.87) to eliminate the $\ddot{\theta}$ term, we have

$$\frac{d\lambda_1}{d\theta} = -3mg \sin \theta,$$

which integrates to

$$\lambda_1(\theta) = 3mg \cos \theta + C. \quad (22.88)$$

To fix the constant C , we evaluate Eq. (22.88) for $\theta = 0$:

$$-ml\dot{\theta}^2 \Big|_{\theta=0} + mg = 3mg + C,$$

which shows that $C \leq -2mg$, with $C = -2mg$ when the initial velocity $\dot{\theta}(0)$ is zero. Using this value of C (which leads to the largest critical angle), we have

$$\lambda_1(\theta) = mg(3 \cos \theta - 2). \quad (22.89)$$

The particle will stay on the surface as long as the force of constraint is nonnegative, that is, as long as the surface has to push outward on the particle, corresponding to $\lambda_1(\theta) > 0$. From Eq. (22.89) we find that the critical angle, at which $\lambda_1(\theta_c) = 0$, satisfies

$$\cos \theta_c = \frac{2}{3}, \quad \text{or} \quad \theta_c = 48^\circ 11'$$

from the vertical. At or before this angle (neglecting all friction) our particle takes off.

It must be admitted that this result can be obtained more easily by considering a varying centripetal force furnished by the radial component of the gravitational force. The example was chosen to illustrate the use of Lagrange's undetermined multiplier without confusing the reader with a complicated physical system. ■

Example 22.4.3 SCHRÖDINGER WAVE EQUATION

As a final illustration of a constrained minimum, let us find the Euler equations for the quantum mechanical problem of a particle of mass m subject to a potential V ,

$$\delta J = \int \psi^*(\mathbf{r}) H \psi(\mathbf{r}) d^3r, \quad (22.90)$$

with the constraint that ψ is the normalized wave function of a bound state:

$$\int \psi^*(\mathbf{r}) \psi(\mathbf{r}) d^3r = 1. \quad (22.91)$$

Equation (22.90) is a statement that the energy of the system is stationary, with H its Hamiltonian operator

$$H = -\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}). \quad (22.92)$$

In Eq. (22.90) ψ and ψ^* are dependent variables; since they are in principle complex we can treat each as a separate variable; this point was discussed in Chapter 5, footnote 3.

The integrand in Eq. (17.121) involves **second** derivatives, but it is convenient to convert them to first derivatives using Green's theorem, Eq. (3.86):

$$\int \psi^*(\mathbf{r}) \nabla^2 \psi(\mathbf{r}) d^3r = \int_S \psi^* \nabla \psi \cdot d\boldsymbol{\sigma} - \int \nabla \psi^* \cdot \nabla \psi d^3r.$$

We now observe that the surface terms vanish due to the requirement that ψ be continuous, and our variational principle becomes

$$\delta \int \left[\frac{\hbar^2}{2m} \nabla \psi^* \cdot \nabla \psi + V \psi^* \psi \right] d^3r = 0. \quad (22.93)$$

The function g for our constrained variation is therefore

$$\begin{aligned} g &= \frac{\hbar^2}{2m} \nabla \psi^* \cdot \nabla \psi + V \psi^* \psi - \lambda \psi^* \psi \\ &= \frac{\hbar^2}{2m} (\psi_x^* \psi_x + \psi_y^* \psi_y + \psi_z^* \psi_z) + V \psi^* \psi - \lambda \psi^* \psi, \end{aligned} \quad (22.94)$$

again using the subscript x to denote $\partial/\partial x$. For $y_i = \psi^*$, our Euler equation becomes

$$\frac{\partial g}{\partial \psi^*} - \frac{\partial}{\partial x} \frac{\partial g}{\partial \psi_x^*} - \frac{\partial}{\partial y} \frac{\partial g}{\partial \psi_y^*} - \frac{\partial}{\partial z} \frac{\partial g}{\partial \psi_z^*} = 0.$$

This yields

$$V \psi - \lambda \psi - \frac{\hbar^2}{2m} (\psi_{xx} + \psi_{yy} + \psi_{zz}) = 0,$$

or

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V \psi = \lambda \psi. \quad (22.95)$$

The Euler equation for $y_i = \psi$ gives the complex conjugate of Eq. (22.95), and therefore provides no further information. Reference to Eq. (22.92) enables us to identify λ physically as the energy of the quantum mechanical system. With this interpretation, Eq. (22.95) is the celebrated Schrödinger wave equation. ■

Rayleigh-Ritz Technique

A number of physically important problems can be related to variational principles of the general form

$$\delta J = \delta \int_a^b \left(p(x)y_x^2 + q(x)y^2 \right) dx = 0, \quad (22.96)$$

where $y(a)$ and $y(b)$ have fixed values, and the variation is subject to the constraint

$$\int_a^b y^2 w(x) dx = \text{constant}. \quad (22.97)$$

Treating Eqs. (22.96) and (22.97) as a constrained minimization, its Euler equation takes the form

$$\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) - q(x)y + \lambda w y = 0, \quad (22.98)$$

where λ is a Lagrange multiplier. This situation usually arises in contexts such that $w(x)$ is a nonnegative weight function and $y(a)$ and $y(b)$ satisfy Sturm-Liouville boundary conditions, meaning that

$$p(x)y_x y \Big|_a^b = 0. \quad (22.99)$$

From the above we conclude that although originally introduced as a Lagrange multiplier, λ must also be an eigenvalue of the Sturm-Liouville system described by Eqs. (22.98) and (22.99). This identification was already noted in Example 22.4.3.

Often problems of the type now under discussion are presented as unconstrained minimizations of the form

$$\delta J = \delta \left(\frac{\int_a^b \left(p(x)y_x^2 + q(x)y^2 \right) dx}{\int_a^b y^2 w(x) dx} \right) = 0. \quad (22.100)$$

Equation (22.100) is equivalent to the earlier formulation because $py_x^2 + qy^2$ is homogeneous in y and the denominator normalizes y without otherwise changing its functional form. The J satisfying Eq. (22.100) evaluates to the eigenvalue λ .

In the frequently occurring case that $p(x)$ is actually independent of x , we can manipulate the y_x^2 term in the integrand of J , causing Eqs. (22.96), (22.97), and (22.100) to assume the useful forms,

$$\delta J = \delta \int_a^b (-p y_{xx} + q(x)y^2) dx = 0, \quad (\text{constrained minimum}), \quad (22.101)$$

$$p \frac{d^2 y}{dx^2} - q(x)y + \lambda w y = 0, \quad (22.102)$$

$$\delta J = \delta \left(\frac{\int_a^b (-p y y_{xx} + q(x)y^2) dx}{\int_a^b y^2 w(x) dx} \right) = 0, \quad (\text{unconstrained, } J = \lambda). \quad (22.103)$$

The Rayleigh-Ritz technique uses the direct evaluation of any one of the above forms for $\delta J = 0$ as a means of obtaining solutions to the eigenvalue problem shown as Eq. (22.98) or (22.102). Application of the technique can be as simple as guessing a form for y and evaluating J , but more accurate results are obtained by taking a form for $y(x)$ that contains adjustable parameters, and then varying the parameters to minimize J within the parameter space. The quality of the results obtained obviously depends on whether the actual minimum form for y has been well approximated.

Ground State Eigenfunction

Suppose that we seek to compute the ground-state eigenfunction y_0 and eigenvalue λ_0 of some complicated atomic or nuclear system.⁹ A classic example, for which no exact analytical solution has been found, is the helium atom problem. The eigenfunction y_0 is **unknown**, but we shall assume that we can make a pretty good guess at an approximation to it, which we will call y . Although we do not know either y_0 or any other eigenfunctions y_i ($i = 1, 2, \dots$), or the corresponding eigenvalues λ_i , we do know, because the eigenfunctions can be chosen to form a complete orthogonal set, that we can write the expansion

$$y = c_0 y_0 + \sum_{i=1}^{\infty} c_i y_i. \quad (22.104)$$

We shall assume that we picked y sensibly enough that it is not orthogonal to the ground state, so $c_0 \neq 0$. Invoking the orthogonality property, E_y , the expectation value of the

⁹This means that λ_0 is the smallest eigenvalue.

energy for wave function y , is

$$E_y = \frac{\langle y | H | y \rangle}{\langle y | y \rangle} = \frac{\sum_{i=0}^{\infty} |c_i|^2 \lambda_i}{\sum_{i=0}^{\infty} |c_i|^2}, \quad (22.105)$$

where H , the operator defining the Schrödinger equation, typically has the form

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x).$$

The Schrödinger equation and its approximate solution E_y are then seen to correspond to Eqs. (22.102) and (22.103). The final member of Eq. (22.105) results from the substitution of Eq. (22.104). This substitution is similar to that carried out in Eq. (6.30), but note that in that equation the function (there called ψ) was assumed normalized. As we already observed in Section 6.4, the expression for E_y is a weighted average of the eigenvalues (with all the weights $|c_i|^2 \geq 0$), so E_y must be at least as large as y_0 , and in fact must be larger if y contains any admixture of eigenfunctions whose λ_i are larger than λ_0 .

It is useful to scale y so $c_0 = 1$ and rearrange Eq. (22.105) to

$$E_y = \lambda_0 + \frac{\sum_{i=1}^{\infty} c_i^2 \lambda_i}{1 + \sum_{i=1}^{\infty} c_i^2}, \quad (22.106)$$

a form that makes clear that the error in E_y will be quadratic in the c_i , even though the difference between y and y_0 is linear in the c_i .

Our analysis therefore contains two important results.

- (1) *Whereas the error in the eigenfunction y was $\mathcal{O}(c_i)$, the error in λ is only $\mathcal{O}(c_i^2)$. Even a poor approximation of the eigenfunctions may yield an accurate calculation of the eigenvalue.*
- (2) *If λ_0 is the lowest eigenvalue (ground state), then $E_y > \lambda_0$, so our approximation is always on the high side, but converges to λ_0 as our approximate eigenfunction y improves ($c_i \rightarrow 0$).*

In practical problems in quantum mechanics, y often depends on parameters that may be varied to minimize E_y and thereby improve the estimate of the ground-state energy λ_0 . This is the “variational method” discussed in quantum mechanics texts. It was illustrated in Example 8.4.1.

Example 22.4.4 QUANTUM OSCILLATOR

The ground state of a quantum-mechanical particle of mass m constrained to the region $0 \leq x < \infty$ and subject also to a potential $V = kx^2/2$ is described (in a unit system with

$\hbar = 1$) by the lowest-eigenvalue eigenstate of the Schrödinger equation,

$$-\frac{1}{2m} \frac{d^2\psi}{dx^2} + \frac{kx^2}{2} \psi = E\psi, \quad (22.107)$$

subject to the boundary conditions $\psi(0) = \psi(\infty) = 0$. A guessed wave function consistent with the boundary conditions is $y(x) = xe^{-\alpha x}$. Let's find the value of α making the approximate eigenvalue a minimum.

Our Schrödinger equation is of the type represented by Eq. (22.102), so we can use Eq. (22.103) and find the unconstrained value of J as given there with $p = 1/2m$, $q = kx^2/2$, $w = 1$, and integration range $(0, \infty)$. Noting that $y_{xx} = \alpha(\alpha x - 2)e^{-\alpha x}$, we have

$$J = \frac{\int_0^\infty \left(-\frac{\alpha x}{2m}(\alpha x - 2) + \frac{kx^4}{2} \right) e^{-2\alpha x} dx}{\int_0^\infty x^2 e^{-2\alpha x} dx} = \frac{\frac{1}{8m\alpha} + \frac{3k}{8\alpha^5}}{\frac{1}{4\alpha^3}} = \frac{\alpha^2}{2m} + \frac{3k}{2\alpha^2}. \quad (22.108)$$

Differentiating Eq. (22.108) with respect to α^2 and setting the result to zero, we get

$$\frac{1}{2m} - \frac{3k}{2\alpha^4} = 0, \quad \text{or} \quad \alpha = (3mk)^{1/4}.$$

Inserting this α value into the expression for J , Eq. (22.108), we find

$$J = \frac{(3mk)^{1/2}}{2m} + \frac{3k}{2(3mk)^{1/2}} = \sqrt{\frac{3k}{m}} \approx 1.732\sqrt{\frac{k}{m}}. \quad (22.109)$$

This value of J is an upper bound to the ground-state energy, the exact value of which is $1.5\sqrt{k/m}$.

Taking a somewhat more complicated (and flexible) wave function, of the form $y = (x + cx^2)e^{-\alpha x}$, and optimizing both α and c , the approximate energy improves to $1.542\sqrt{k/m}$. The approximate wave functions of this example are compared with the exact wave function in Fig. 22.11. Note that the second approximation yields an eigenvalue that is in error

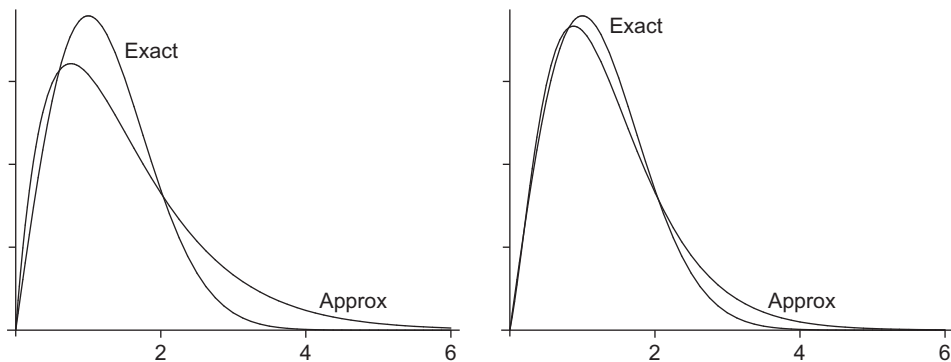


FIGURE 22.11 Exact and approximate ground-state wave functions for quantum oscillator, Example 22.4.4, plotted for $k/m = 1$. Left: Single-term approximation $y = xe^{-\alpha x}$. Right: Two-term approximation $y = (x + cx^2)e^{-\alpha x}$.

by less than 3%, even though the approximate wave function exhibits considerably larger relative errors. ■

Example 22.4.5 VARIATION OF LINEAR PARAMETERS

A frequent use of the Rayleigh-Ritz technique is the approximation of an eigenfunction of a Schrödinger equation,

$$H\psi(x) = E\psi(x),$$

as a truncated expansion in a fixed orthonormal set of functions. The advantage of this procedure is that the parameters in the wave function all occur linearly, and the optimization reduces to a matrix eigenvalue problem.

Given an approximate function (often called a **trial function**) of the form

$$y(x) = \sum_{i=1}^N c_i \varphi_i(x), \quad (22.110)$$

we seek to minimize

$$J = \langle y | H | y \rangle \quad \text{subject to} \quad \langle y | y \rangle = 1. \quad (22.111)$$

Again we emphasize that the φ_i have no specific relation to the eigenfunction we seek; they are simply members of an orthonormal set that has two desirable features: (1) they are such that a few of them can provide a reasonable representation of the eigenfunction, and (2) they are tractable in the sense that it is convenient to evaluate the matrix elements we are about to define.

Defining a matrix $\mathbf{H}$ of elements $H_{ij} = \langle \varphi_i | H | \varphi_j \rangle$ and a column vector $\mathbf{c}$ with components c_i , we can restate Eq. (22.111) as the minimization of

$$J = \mathbf{c}^\dagger \mathbf{H} \mathbf{c} \quad \text{subject to} \quad \mathbf{c}^\dagger \mathbf{c} = 1. \quad (22.112)$$

This formulation, in turn, can be reduced using Lagrangian multipliers to the unconstrained matrix eigenvalue problem

$$\mathbf{H} \mathbf{c} = \lambda \mathbf{c}, \quad (22.113)$$

which we can solve (using matrix methods) for λ (the approximate value of J). This application of the Rayleigh-Ritz technique is therefore seen to be equivalent to the approximation of an operator equation by a finite matrix equation. ■

Exercises

- 22.4.1** A particle of mass m is on a frictionless horizontal surface. In terms of plane polar coordinates (r, θ) , it is constrained to move so that $\theta = \omega t$ (accomplished by pushing it with a rotating radial arm against which it can slide frictionlessly). With the initial conditions

$$t = 0, \quad r = r_0, \quad \dot{r} = 0:$$

- (a) Find the radial position as a function of time.

$$\text{ANS. } r(t) = r_0 \cosh \omega t.$$

- (b) Find the force exerted on the particle by the constraint.

$$\text{ANS. } F^{(c)} = 2m\dot{r}\omega = 2mr_0\omega^2 \sinh \omega t.$$

22.4.2 A point mass m is moving on a flat, horizontal, frictionless plane. The mass is constrained by a string to move radially inward at a constant rate. Using plane polar coordinates (r, θ) , $r = r_0 - kt$:

- Set up the Lagrangian.
- Obtain the constrained Lagrange equations.
- Solve the θ -dependent Lagrange equation to obtain $\omega(t)$, the angular velocity. What is the physical significance of the constant of integration that you get from your “free” integration?
- Using the $\omega(t)$ from part (b), solve the r -dependent (constrained) Lagrange equation to obtain $\lambda(t)$. In other words, explain what is happening to the **force** of constraint as $r \rightarrow 0$.

22.4.3 A flexible cable is suspended from two fixed points. The length of the cable is fixed. Find the curve that will minimize the total gravitational potential energy of the cable.

$$\text{ANS. } \text{Hyperbolic cosine.}$$

22.4.4 A fixed volume of water is rotating in a cylinder with constant angular velocity ω . Find the curve of the water surface that will minimize the total potential energy of the water in the combined gravitational-centrifugal force field.

$$\text{ANS. } \text{Parabola.}$$

- 22.4.5** (a) Show that for a fixed-length perimeter the plane figure with maximum area is a circle.
 (b) Show that for a fixed planar area the boundary with minimum perimeter is a circle.

Hint. The radius of curvature R is given by

$$R = \frac{(r^2 + r_\theta^2)^{3/2}}{rr_{\theta\theta} - 2r_\theta^2 - r^2}.$$

Note. The problems of this section, variation subject to constraints, are often called *isoperimetric*. The term arose from problems of maximizing area subject to a fixed perimeter, as in part (a) of this problem.

22.4.6 Show that requiring J , given by

$$J = \int_a^b \int_a^b K(x, t) \varphi(x) \varphi(t) dx dt,$$

to have a stationary value subject to the normalizing condition

$$\int_a^b \varphi^2(x) dx = 1$$

leads to a Hilbert-Schmidt integral equation of the form given in Eq. (20.64).

Note. The kernel $K(x, t)$ is symmetric.

22.4.7 An unknown function satisfies the differential equation

$$y'' + \left(\frac{\pi}{2}\right)^2 y = 0$$

and the boundary conditions $y(0) = 1$, $y(1) = 0$.

- (a) Calculate the approximation $\lambda = F[y_{\text{trial}}]$ for $y_{\text{trial}} = 1 - x^2$.
- (b) Compare with the exact eigenvalue.

ANS. (a) $\lambda = 2.5$. (b) $\lambda/\lambda_{\text{exact}} = 1.013$.

22.4.8 In [Exercise 22.4.7](#) use a trial function $y = 1 - x^n$.

- (a) Find the value of n that will minimize $F[y_{\text{trial}}]$.
- (b) Show that the optimum value of n drives the ratio $\lambda/\lambda_{\text{exact}}$ down to 1.003.

ANS. (a) $n = 1.7247$.

22.4.9 A quantum mechanical particle in a sphere (Example 14.7.1) satisfies

$$\nabla^2 \psi + k^2 \psi = 0,$$

with $k^2 = 2mE/\hbar^2$. The boundary condition is that $\psi = 0$ at $r = a$, where a is the radius of the sphere. For the ground state [where $\psi = \psi(r)$], try an approximate wave function,

$$\psi_a(r) = 1 - \left(\frac{r}{a}\right)^2,$$

and calculate an approximate eigenvalue k_a^2 .

Hint. To determine $p(r)$ and $w(r)$, put your equation in self-adjoint form (in spherical polar coordinates).

ANS. $k_a^2 = \frac{10.5}{a^2}$, $k_{\text{exact}}^2 = \frac{\pi^2}{a^2}$.

22.4.10 The wave equation for a quantum mechanical oscillator may be written as

$$\frac{d^2 \psi(x)}{dx^2} + (\lambda - x^2) \psi(x) = 0,$$

with $\lambda = 1$ for the ground state; see Eq. (18.17). Take

$$\psi_{\text{trial}} = \begin{cases} 1 - \frac{x^2}{a^2}, & x^2 \leq a^2 \\ 0, & x^2 > a^2 \end{cases}$$

for the ground-state wave function (with a^2 an adjustable parameter) and calculate the corresponding ground-state energy. How much error do you have?

Note. Your parabola is really not a very good approximation to a Gaussian exponential. What improvements can you suggest?

22.4.11 The Schrödinger equation for a central potential may be written as

$$\mathcal{L}u(r) + \frac{\hbar^2 l(l+1)}{2Mr^2} u(r) = Eu(r).$$

The $l(l+1)$ term, the angular momentum barrier, comes from splitting off the angular dependence. Compare Eq. (9.80) (divide that equation through by $-r^2$). Use the Rayleigh-Ritz technique to show that $E > E_0$, where E_0 is the energy eigenvalue of $\mathcal{L}u_0 = E_0 u_0$ corresponding to $l = 0$. This means that the ground state will have $l = 0$, zero angular momentum.

Hint. You can expand $u(r)$ as $u_0(r) + \sum_{i=1}^{\infty} c_i u_i$, where $\mathcal{L}u_i = E_i u_i$, $E_i > E_0$.

Additional Readings

Bliss, G. A., *Calculus of Variations*. The Mathematical Association of America. LaSalle, IL: Open Court Publishing Co. (1925). As one of the older texts, this is still a valuable reference for details of problems such as minimum-area problems.

Courant, R., and H. Robbins, *What Is Mathematics?* 2nd ed. New York: Oxford University Press (1996). Chapter VII contains a fine discussion of the calculus of variations, including soap-film solutions to minimum-area problems.

Ewing, G. M., *Calculus of Variations with Applications*. New York: Norton (1969). Includes a discussion of sufficiency conditions for solutions of variational problems.

Lanczos, C., *The Variational Principles of Mechanics*, 4th ed. Toronto: University of Toronto Press (1970), reprinted, Dover (1986). This book is a very complete treatment of variational principles and their applications to the development of classical mechanics.

Sagan, H., *Boundary and Eigenvalue Problems in Mathematical Physics*. New York: Wiley (1961), reprinted, Dover (1989). This delightful text could also be listed as a reference for Sturm-Liouville theory, Legendre and Bessel functions, and Fourier series. Chapter 1 is an introduction to the calculus of variations, with applications to mechanics. Chapter 7 picks up the calculus of variations again and applies it to eigenvalue problems.

Sagan, H., *Introduction to the Calculus of Variations*. New York: McGraw-Hill (1969), reprinted, Dover (1983). This is an excellent introduction to the modern theory of the calculus of variations, which is more sophisticated and complete than his 1961 text. Sagan covers sufficiency conditions and relates the calculus of variations to problems of space technology.

Weinstock, R., *Calculus of Variations*. New York: McGraw-Hill (1952), New York: Dover (1974). A detailed, systematic development of the calculus of variations and applications to Sturm-Liouville theory and physical problems in elasticity, electrostatics, and quantum mechanics.

Yourgrau, W., and S. Mandelstam, *Variational Principles in Dynamics and Quantum Theory*, 3rd ed. Philadelphia: Saunders (1968), New York: Dover (1979). This is a comprehensive, authoritative treatment of variational principles. The discussions of the historical development and the many metaphysical pitfalls are of particular interest.